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(54) **SEMICONDUCTOR PACKAGE**
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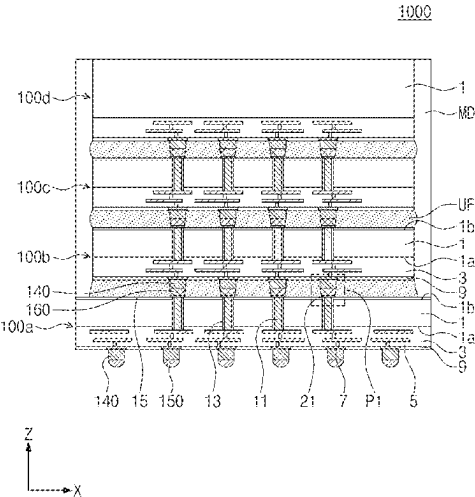
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25/0657
See application file for complete search history.

(57) **ABSTRACT**

A semiconductor package includes a sequential stack of first
and second semiconductor chips, and a first internal con-
nection member that connects the first and second semicon-
ductor chips to each other. The first semiconductor chip
includes a first substrate that has a first top surface and a first
bottom surface that are opposite to each other, and a first
conductive pad on the first top surface. The second semi-
conductor chip includes a second substrate that has a second
top surface and a second bottom surface that are opposite to
each other, and a second conductive bump on the second
bottom surface. The first internal connection member con-
nects the first conductive pad to the second conductive
bump. The first conductive pad has a first width in one
direction. The second conductive bump has a second width
in the one direction. The first width is smaller than the
second width.

20 Claims, 14 Drawing Sheets



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2224/16013 (2013.01); *H01L 2224/81203*
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(2013.01); *H01L 2225/1058* (2013.01); *H01L*
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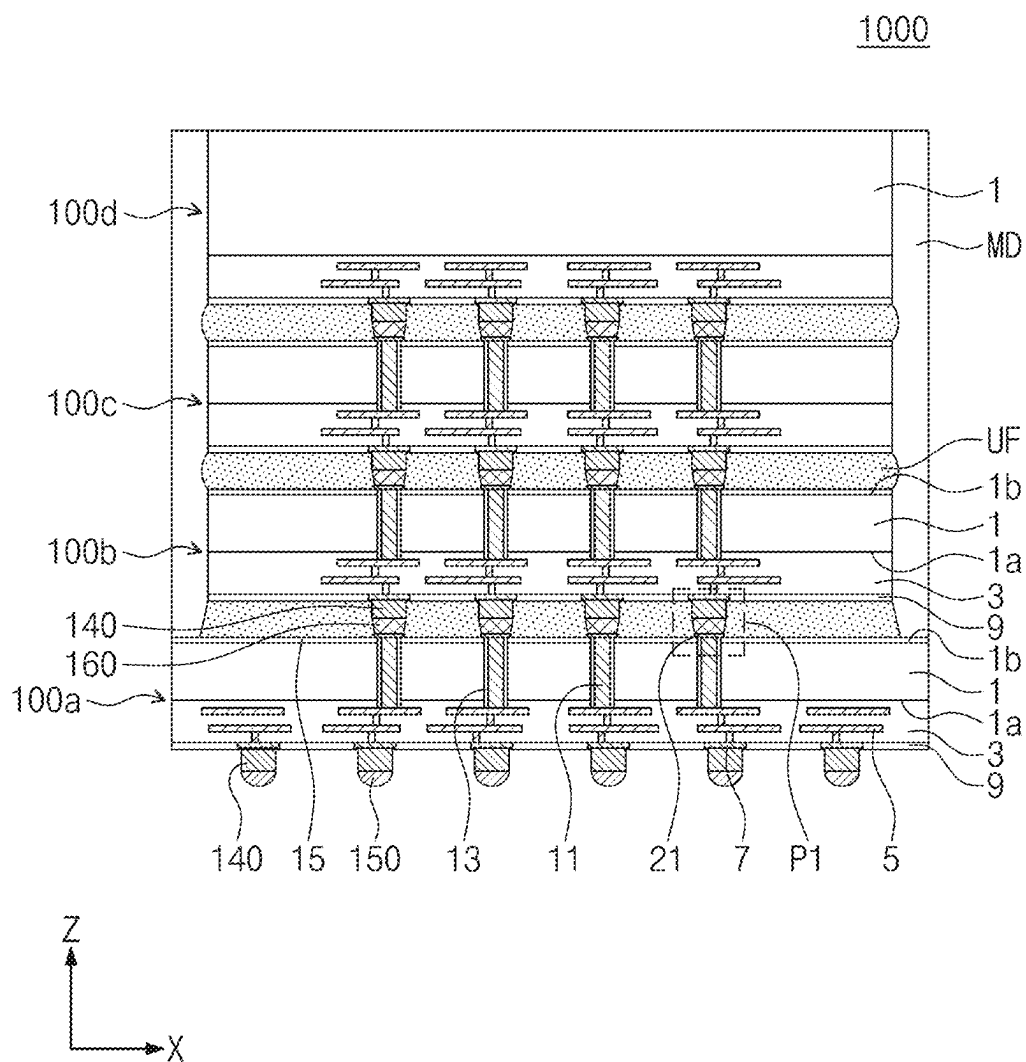


FIG. 2A

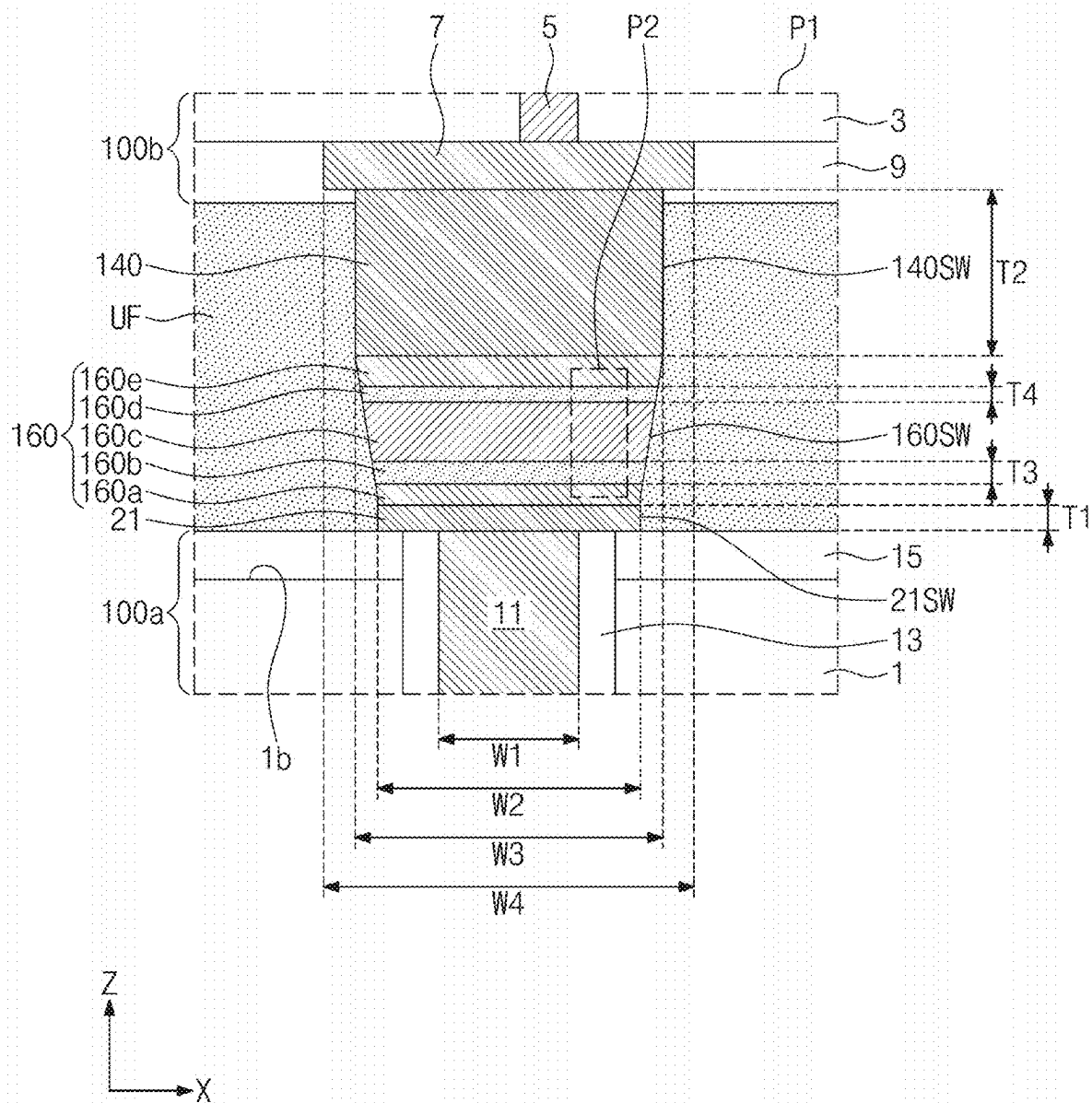


FIG. 2B

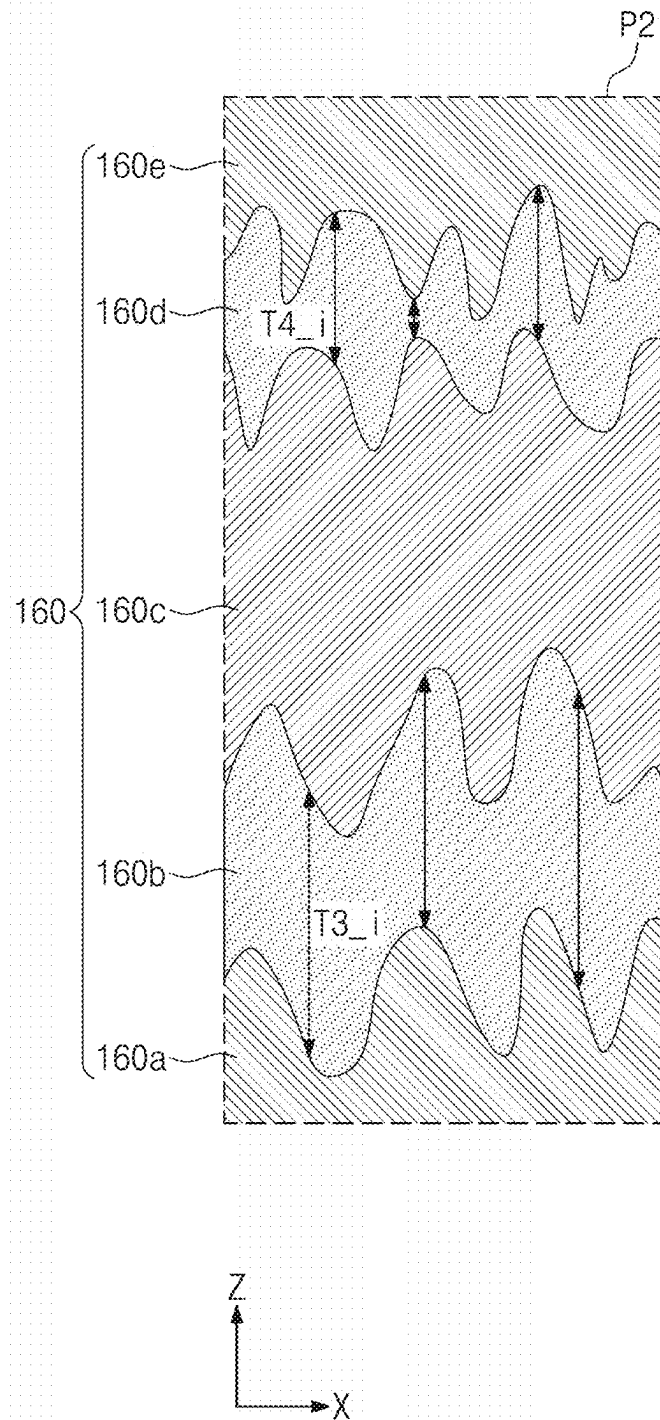


FIG. 2C

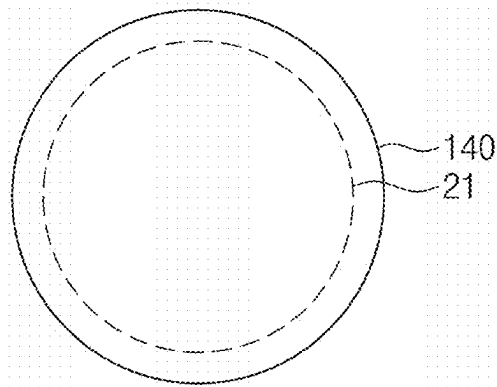
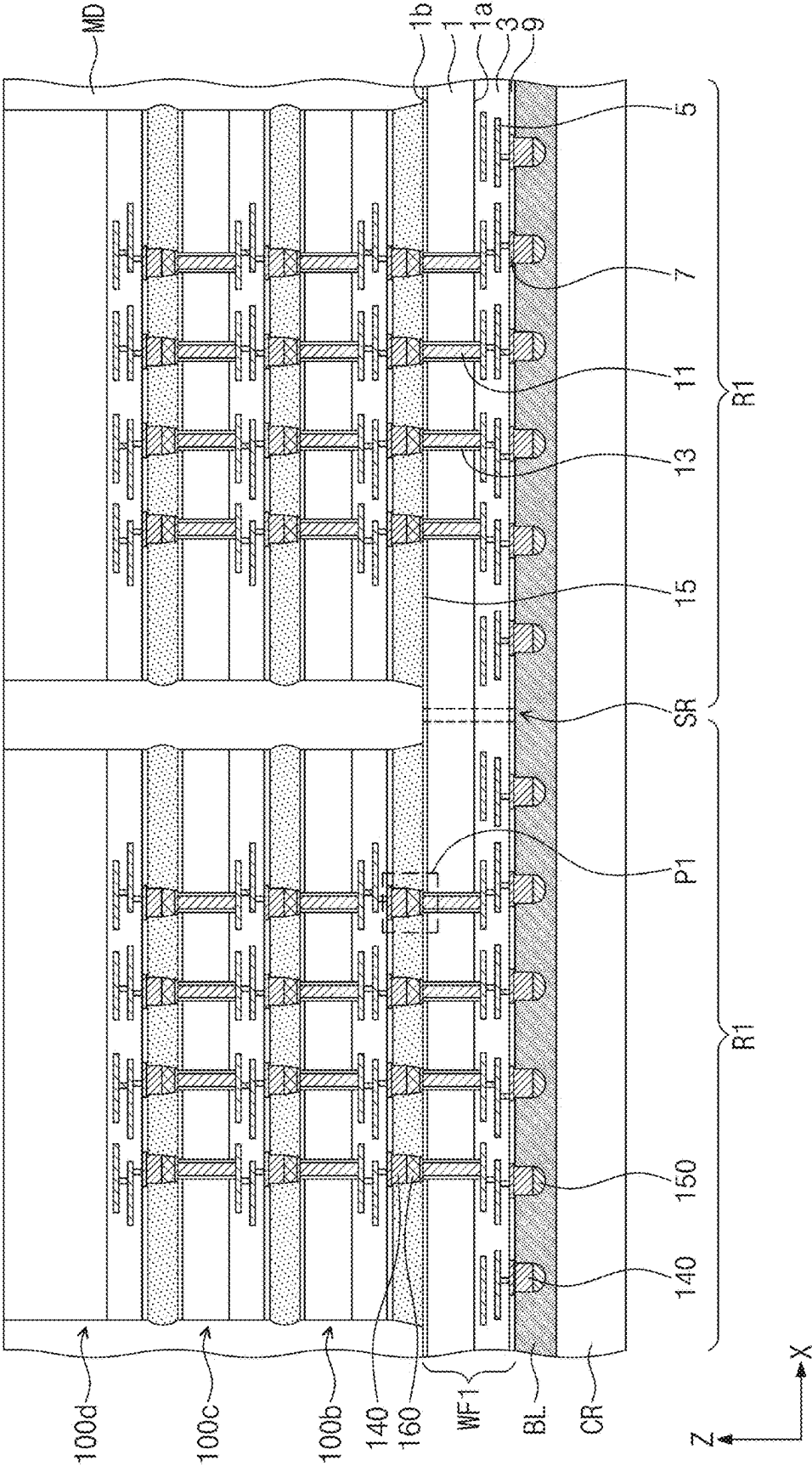
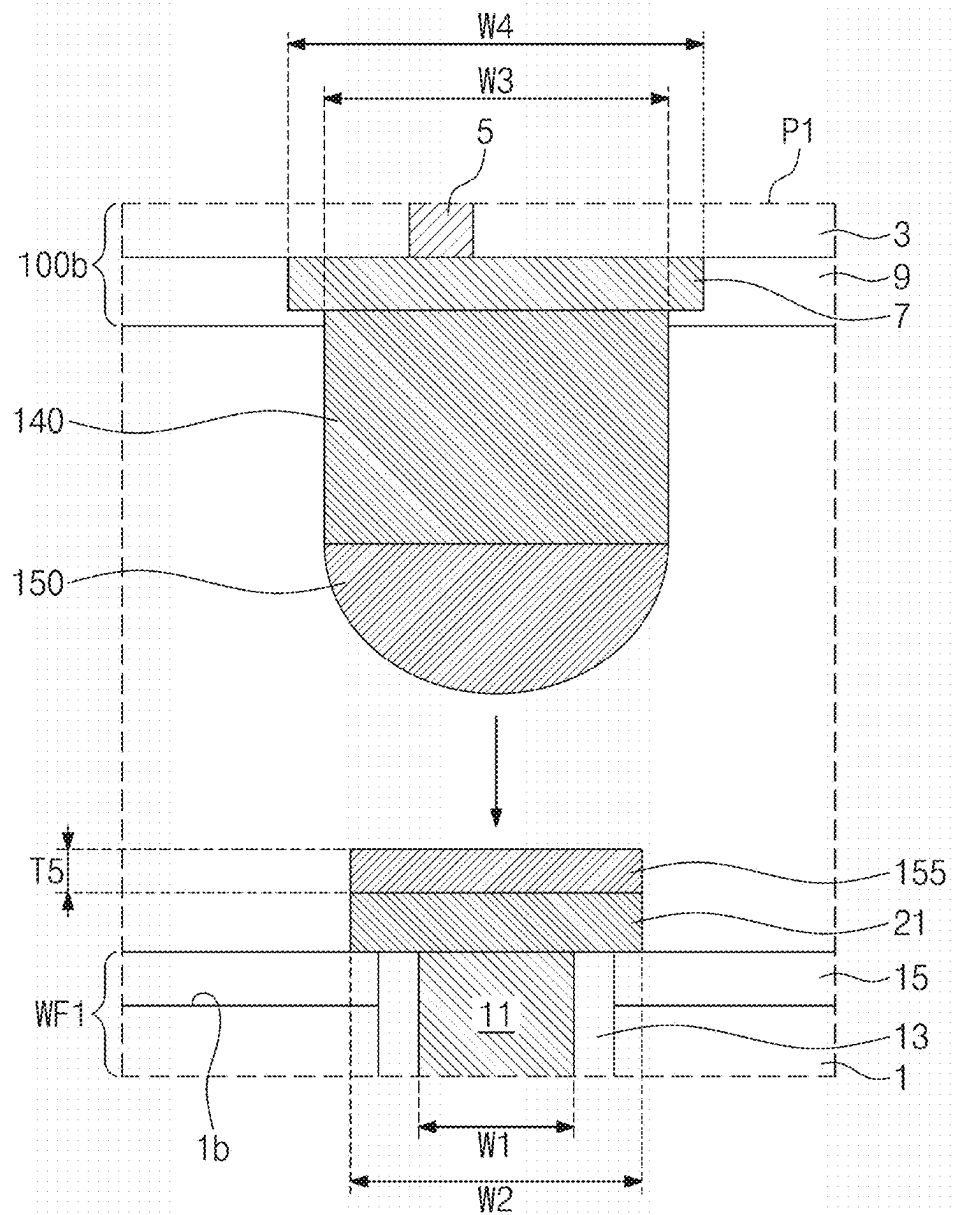


FIG. 3B





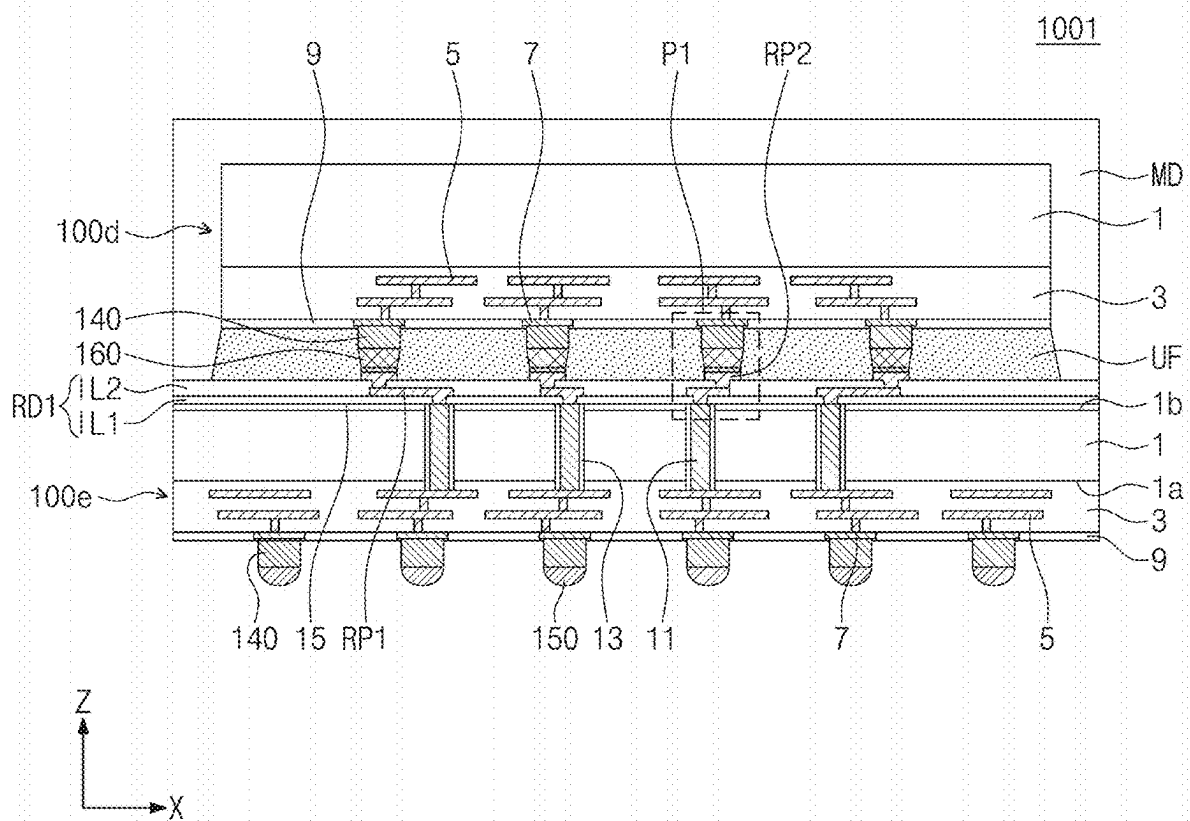


FIG. 6

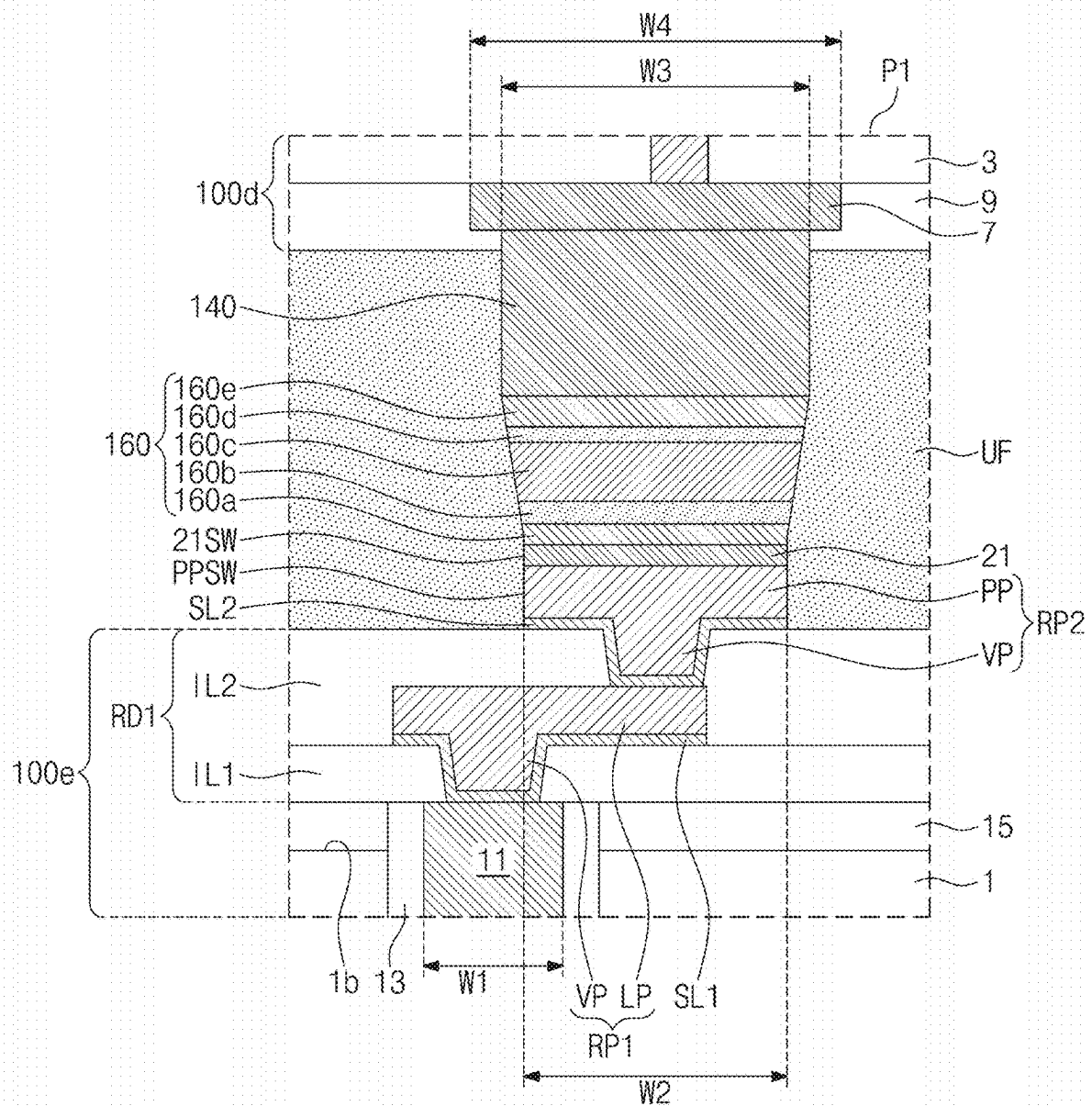


FIG. 7

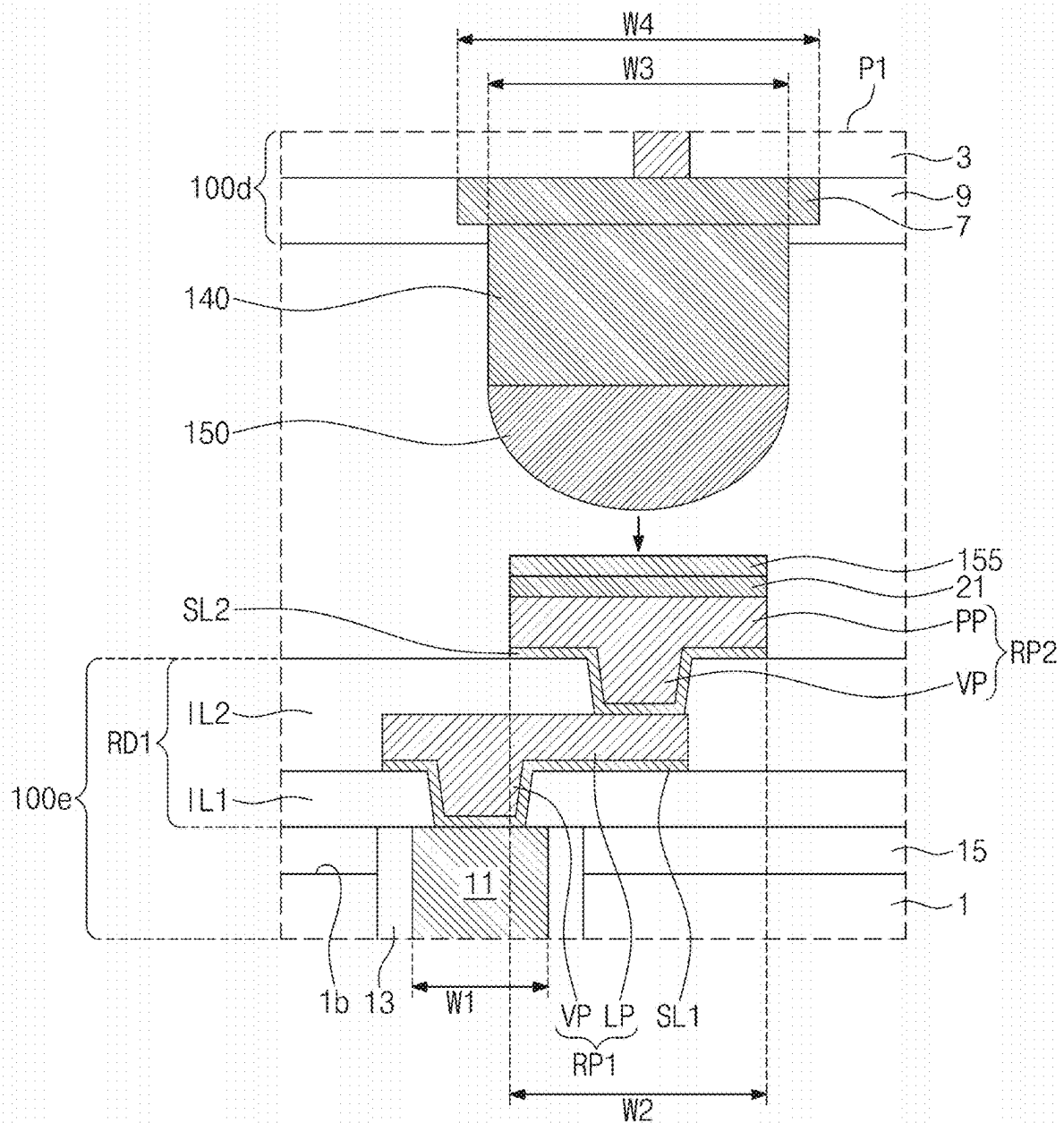


FIG. 8

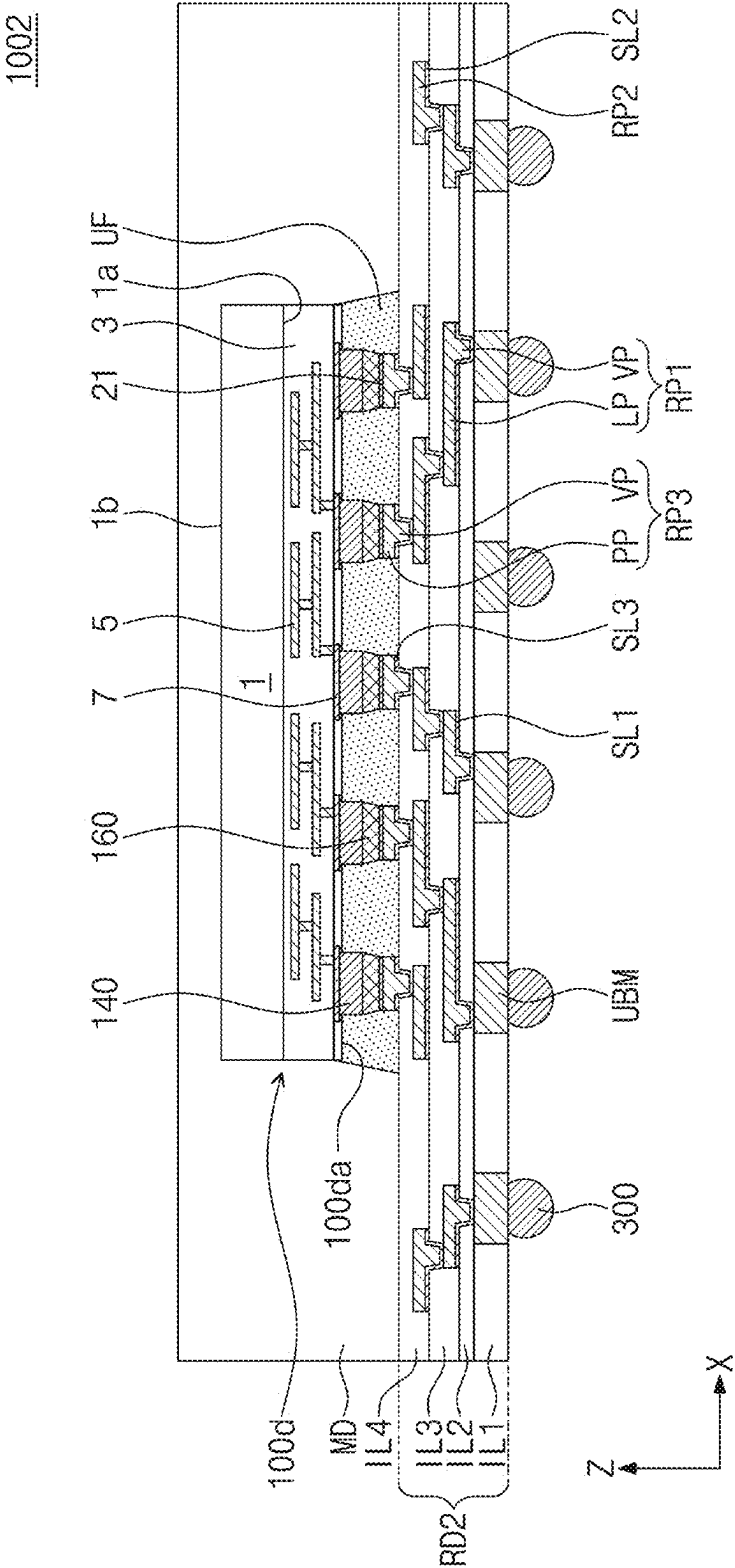


FIG. 9

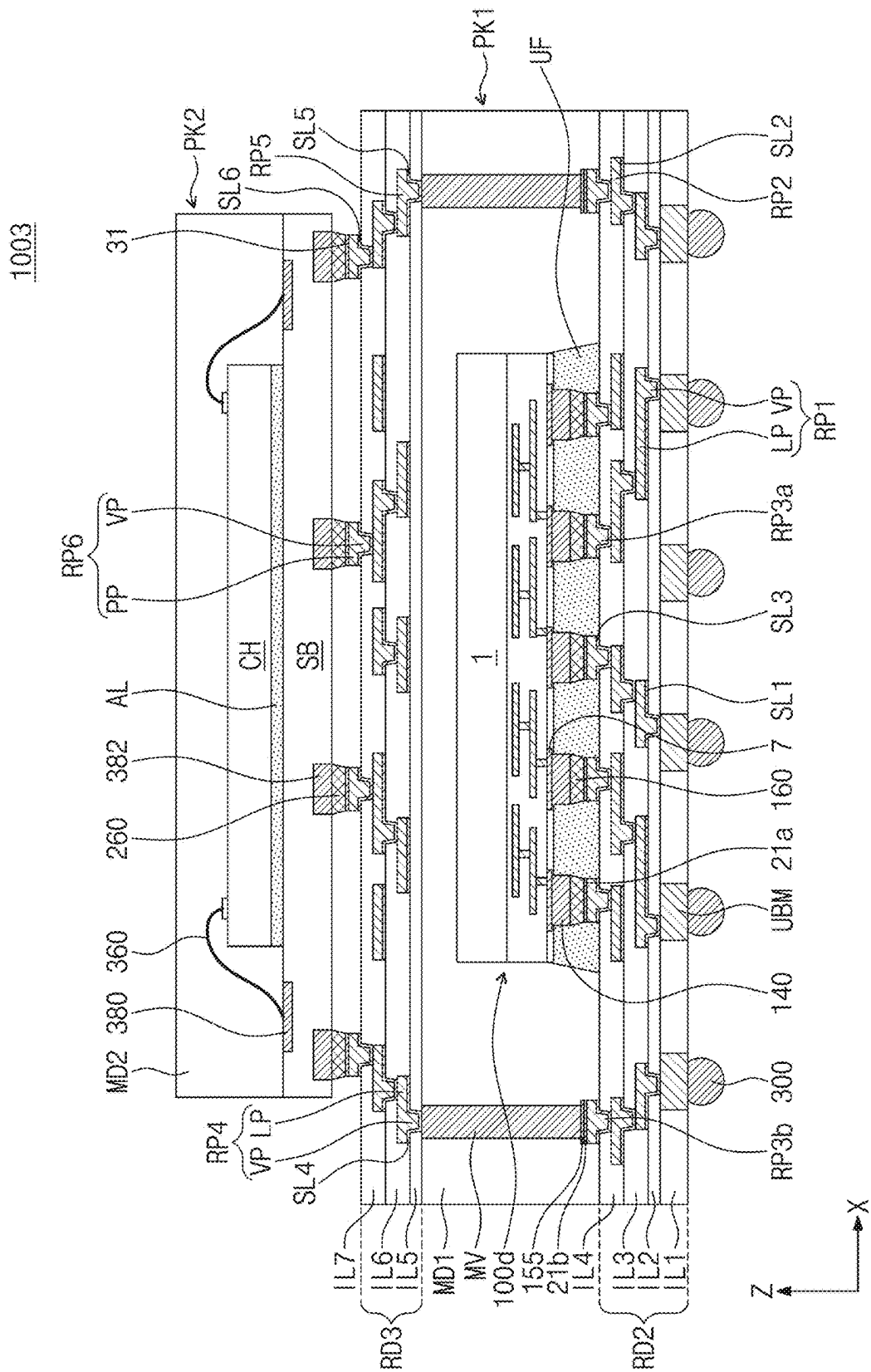
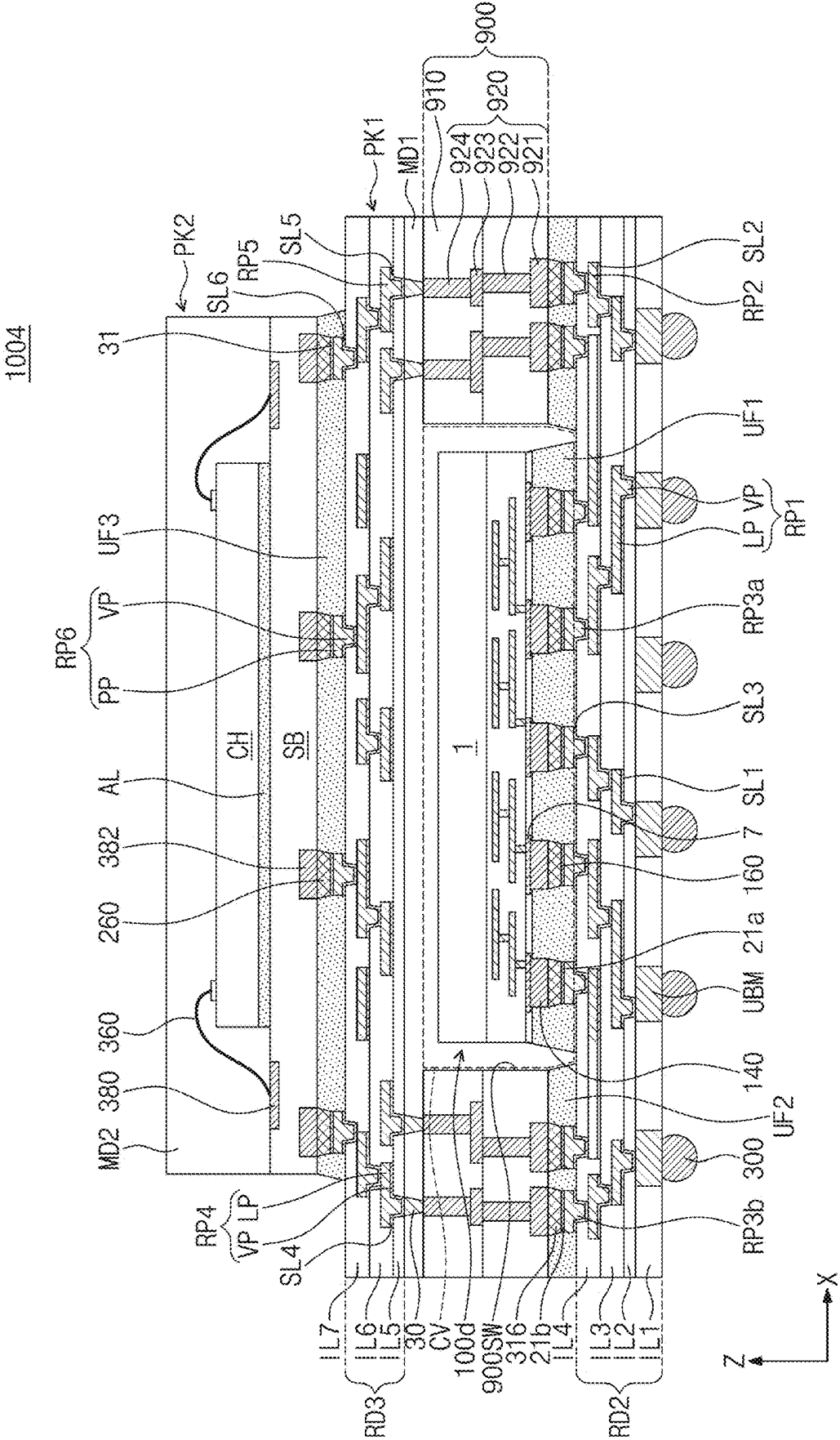
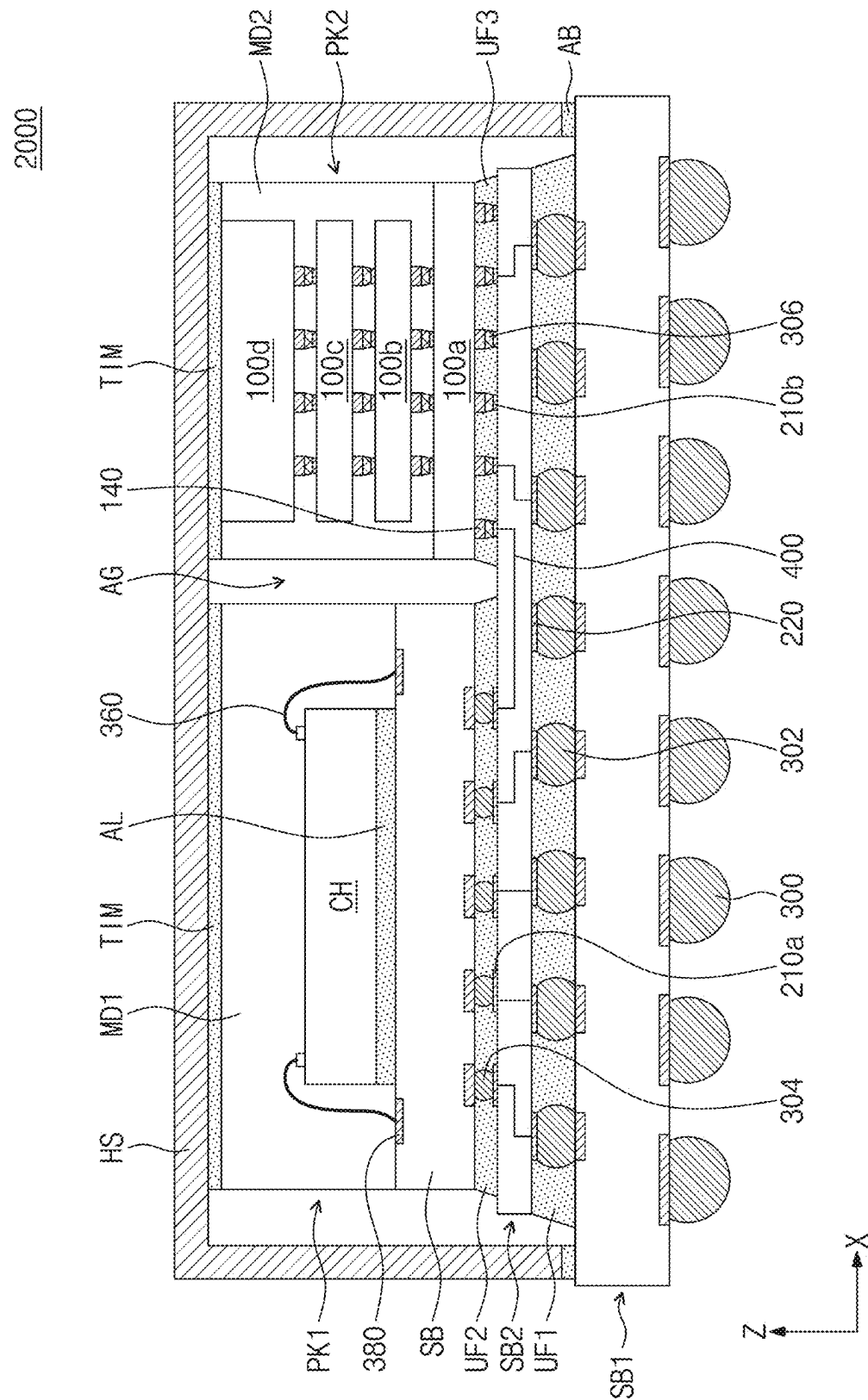


FIG. 10



III



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SEMICONDUCTOR PACKAGE**CROSS-REFERENCE TO RELATED APPLICATIONS**

This U.S. nonprovisional application is a continuation application of U.S. application Ser. No. 17,376,616, dated Jul. 15, 2021, which claims priority under 35 U.S.C. § 119 to Korean Patent Application No. 10-2020-0122493 filed on Sep. 22, 2020 in the Korean Intellectual Property Office, the disclosure of each of which is hereby incorporated by reference in its entirety.

BACKGROUND

The present inventive concepts relate to semiconductor packages.

A semiconductor package is provided to implement an integrated circuit chip to qualify for use in electronic products. A semiconductor package is typically configured such that a semiconductor chip is mounted on a printed circuit board (PCB) and bonding wires or bumps are used to electrically connect the semiconductor chip to the printed circuit board. With the development of electronic industry, many studies have been conducted to improve reliability and durability of semiconductor packages.

SUMMARY

Some example embodiments of the present inventive concepts provide semiconductor packages with increased reliability.

An object of the present inventive concepts is not limited to the mentioned above, and other objects which have not been mentioned above will be clearly understood to those skilled in the art from the following description.

According to some example embodiments of the present inventive concepts, a semiconductor package may comprise: a sequential stack of a first semiconductor chip and a second semiconductor chip; and a first internal connection member that connects the first semiconductor chip to the second semiconductor chip. The first semiconductor chip may include: a first substrate that has a first top surface and a first bottom surface that are opposite to each other; and a first conductive pad on the first top surface. The second semiconductor chip may include: a second substrate that has a second top surface and a second bottom surface that are opposite to each other; and a second conductive bump on the second bottom surface. The first internal connection member may connect the first conductive pad to the second conductive bump. The first conductive pad may have a first width in one direction. The second conductive bump may have a second width in the one direction. The first width may be smaller than the second width.

According to some example embodiments of the present inventive concepts, a semiconductor package may comprise: a sequential stack of a first semiconductor chip and a second semiconductor chip; and a first internal connection member that connects the first semiconductor chip to the second semiconductor chip. The first semiconductor chip may include: a first substrate that has a first top surface and a first bottom surface that are opposite to each other; and a first conductive pad on the first top surface. The second semiconductor chip may include: a second substrate that has a second top surface and a second bottom surface that are opposite to each other; a second conductive pad adjacent to the second bottom surface; and a second conductive bump

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bonded to the second conductive pad. The first internal connection member may connect the first conductive pad to the second conductive bump. The first internal connection member may include: a first connection region adjacent to the first conductive pad; and a second connection region adjacent to the second conductive bump. Each of the first and second connection regions may include gold (Au). The first connection region has a first average thickness. The second connection region may have a second average thickness smaller than the first average thickness.

According to some example embodiments of the present inventive concepts, a semiconductor package may comprise: a sequential stack of a plurality of semiconductor chips; and a mold layer that covers the semiconductor chips. Each of the semiconductor chips may include: a semiconductor substrate that has a top surface and a bottom surface; an interlayer dielectric layer that covers the bottom surface; a first conductive pad below the interlayer dielectric layer; a conductive bump bonded to the first conductive pad; a through via that penetrates the semiconductor substrate; and a second conductive pad on the top surface and in contact with the through via. The second conductive pad may have a first width in one direction. The conductive bump may have a second width in the one direction. The first width may be smaller than the second width.

According to some example embodiments of the present inventive concepts, a semiconductor package may comprise: a first redistribution substrate; a first semiconductor device mounted through a first internal connection member on the first redistribution substrate; and a first mold layer that covers the first semiconductor device and the first redistribution substrate. The first redistribution substrate may include: a first dielectric layer; an under-bump in the first dielectric layer; a second dielectric layer that covers the first dielectric layer; a first redistribution pattern that includes a first via part and a first line part, the first via part penetrating the second dielectric layer and having a connection with the under-bump, and the first line part protruding onto the second dielectric layer; a third dielectric layer that covers the first redistribution pattern and the second dielectric layer; a second redistribution pattern that includes a second via part and a first pad part, the second via part penetrating the third dielectric layer and having a connection with the first redistribution pattern, and the first pad part protruding onto the third dielectric layer; and a first conductive pad on the second redistribution pattern. The first semiconductor device may include a conductive bump that protrudes outwardly from a bottom surface of the first semiconductor device. The first internal connection member may connect the first conductive pad to the conductive bump. A width of the first conductive pad may be smaller than a width of the conductive bump.

BRIEF DESCRIPTION OF THE DRAWINGS

FIG. 1 illustrates a cross-sectional view showing a semiconductor package according to some example embodiments of the present inventive concepts.

FIG. 2A illustrates an enlarged view showing section P1 of FIG. 1 according to some example embodiments of the present inventive concepts.

FIG. 2B illustrates an enlarged view showing section P2 of FIG. 2A according to some example embodiments of the present inventive concepts.

FIG. 2C illustrates a plan view showing a conductive bump and a second conductive pad according to some example embodiments of the present inventive concepts.

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FIGS. 3A and 3B illustrate cross-sectional views showing a method of fabricating the semiconductor package of FIG. 1, according to some example embodiments of the present inventive concepts.

FIG. 4 illustrates an enlarged view showing section P1 of FIG. 3A according to some example embodiments of the present inventive concepts.

FIG. 5 illustrates a cross-sectional view showing a semiconductor package according to some example embodiments of the present inventive concepts.

FIG. 6 illustrates an enlarged view showing section P1 of FIG. 5 according to some example embodiments of the present inventive concepts.

FIG. 7 illustrates a method of fabricating a semiconductor package having a structure of FIG. 6 according to some example embodiments of the present inventive concepts.

FIG. 8 illustrates a cross-sectional view showing a semiconductor package according to some example embodiments of the present inventive concepts.

FIG. 9 illustrates a cross-sectional view showing a semiconductor package according to some example embodiments of the present inventive concepts.

FIG. 10 illustrates a cross-sectional view showing a semiconductor package according to some example embodiments of the present inventive concepts.

FIG. 11 illustrates a cross-sectional view showing a semiconductor package according to some example embodiments of the present inventive concepts.

DETAILED DESCRIPTION

Some example embodiments of the present inventive concepts will now be described in detail with reference to the accompanying drawings to aid in clearly explaining the present inventive concepts. It will be understood that the same reference numerals are assigned to the same or similar constituent elements throughout the specification.

In some example embodiments, when a certain part with a layer, film, region, plate, etc. is said to be “on” another part, the part may be “above” or “below” the other part. In some example embodiments, when a certain part with a layer, film, region, plate, etc. is said to be “on” another part, the part may be “indirectly on” or “directly on” the other part. When a certain part is said to be “indirectly on” another part, an interposing structure and/or space may be present between the certain part and the other part such that the certain part and the other part are isolated from direct contact with each other. Conversely, when a certain part is said to be “directly on” another part, it means that there is no other part between the certain part and the other part such that the certain part is in direct contact with the other part.

It will be understood that elements and/or properties thereof (e.g., structures, surfaces, directions, or the like), which may be referred to as being “perpendicular,” “parallel,” “coplanar,” or the like with regard to other elements and/or properties thereof (e.g., structures, surfaces, directions, or the like) may be “perpendicular,” “parallel,” “coplanar,” or the like or may be “substantially perpendicular,” “substantially parallel,” “substantially coplanar,” respectively, with regard to the other elements and/or properties thereof.

Elements and/or properties thereof (e.g., structures, surfaces, directions, or the like) that are “substantially perpendicular” with regard to other elements and/or properties thereof will be understood to be “perpendicular” with regard to the other elements and/or properties thereof within manufacturing tolerances and/or material tolerances and/or have a

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deviation in magnitude and/or angle from “perpendicular,” or the like with regard to the other elements and/or properties thereof that is equal to or less than 10% (e.g., a tolerance of $\pm 10\%$).

Elements and/or properties thereof (e.g., structures, surfaces, directions, or the like) that are “substantially parallel” with regard to other elements and/or properties thereof will be understood to be “parallel” with regard to the other elements and/or properties thereof within manufacturing tolerances and/or material tolerances and/or have a deviation in magnitude and/or angle from “parallel,” or the like with regard to the other elements and/or properties thereof that is equal to or less than 10% (e.g., a tolerance of $\pm 10\%$).

Elements and/or properties thereof (e.g., structures, surfaces, directions, or the like) that are “substantially coplanar” with regard to other elements and/or properties thereof will be understood to be “coplanar” with regard to the other elements and/or properties thereof within manufacturing tolerances and/or material tolerances and/or have a deviation in magnitude and/or angle from “coplanar,” or the like with regard to the other elements and/or properties thereof that is equal to or less than 10% (e.g., a tolerance of $\pm 10\%$).

It will be understood that elements and/or properties thereof may be recited herein as being “the same” or “equal” as other elements, and it will be further understood that elements and/or properties thereof recited herein as being “the same” as or “equal” to other elements may be “the same” as or “equal” to or “substantially the same” as or “substantially equal” to the other elements and/or properties thereof. Elements and/or properties thereof that are “substantially the same” as or “substantially equal” to other elements and/or properties thereof will be understood to include elements and/or properties thereof that are the same as or equal to the other elements and/or properties thereof within manufacturing tolerances and/or material tolerances. Elements and/or properties thereof that are the same or substantially the same as other elements and/or properties thereof may be structurally the same or substantially the same, functionally the same or substantially the same, and/or compositionally the same or substantially the same.

It will be understood that elements and/or properties thereof described herein as being the “substantially” the same encompasses elements and/or properties thereof that have a relative difference in magnitude that is equal to or less than 10%. Further, regardless of whether elements and/or properties thereof are modified as “substantially,” it will be understood that these elements and/or properties thereof should be construed as including a manufacturing or operational tolerance (e.g., $\pm 10\%$) around the stated elements and/or properties thereof.

When the terms “about” or “substantially” are used in this specification in connection with a numerical value, it is intended that the associated numerical value include a tolerance of $\pm 10\%$ around the stated numerical value. When ranges are specified, the range includes all values therebetween such as increments of 0.1%.

FIG. 1 illustrates a cross-sectional view showing a semiconductor package according to some example embodiments of the present inventive concepts.

Referring to FIG. 1, a semiconductor package 1000 according to some example embodiments may include first, second, third, and fourth semiconductor chips 100a, 100b, 100c, and 100d that are sequentially stacked, and may also include a mold layer MD that covers the first, second, third, and fourth semiconductor chips 100a, 100b, 100c, and 100d. Semiconductor chips that are described herein as being sequentially stacked may be interchangeably referred to as a

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sequential stack of said semiconductor chips. As shown, the second semiconductor chip **100b** may immediately overlie the first semiconductor chip **100a** in the second direction Z. The first semiconductor chip **100a** may have a width greater (e.g., larger) than those of the second, third, and fourth semiconductor chips **100b**, **100c**, and **100d**. The first semiconductor chip **100a** may be of a different type from the second, third, and fourth semiconductor chips **100b**, **100c**, and **100d**. The first semiconductor chip **100a** may be, for example, a logic circuit chip. The second, third, and fourth semiconductor chips **100b**, **100c**, and **100d** may be of the same type (e.g., DRAM chip). Some example embodiments disclose a structure where one logic circuit chip and three memory chips are stacked, but the number of the logic circuit chip and the number of the memory chips may be variously changed without being limited thereto. The semiconductor package **1000** may have a high bandwidth memory (HBM) chip structure.

Each of the first, second, third, and fourth semiconductor chips **100a**, **100b**, **100c**, and **100d** may include a substrate **1**, an interlayer dielectric layer **3**, wiring lines **5**, first conductive pads **7**, conductive bumps **140**, a first passivation layer **9**, and a second passivation layer **15** and one or more of the first, second, third, and fourth semiconductor chips **100a**, **100b**, **100c**, and **100d** may further include at least second conductive pads **21**. Respective elements of the first, second, third, and fourth semiconductor chips **100a**, **100b**, **100c**, and **100d** may be referred to as respective first, second, third, and fourth elements thereof. For example, the substrate **1** of the first semiconductor chip **100a** may be referred to as a first substrate, one or more of the first and/or second conductive pads **7** and/or **21** of the first semiconductor chip **100a** may be referred to as a first conductive pad, etc.

The substrate **1** of one or more semiconductor chips may have a substrate top surface **1b** and a substrate bottom surface **1a** that are opposite to each other. For example, as shown in FIG. 1, the first semiconductor chip **100a** may include a substrate **1**, referred to as a first substrate, that has a top surface **1b** and a bottom surface **1a** that are opposite to each other, and the second semiconductor chip **100b** may include a substrate **1**, referred to as a second substrate, that has a top surface **1b** and a bottom surface **1a** that are opposite to each other. As described herein, the one or more semiconductor chips may include one or more of the first, second, third, and/or fourth semiconductor chips **100a**, **100b**, **100c**, and/or **100d**. The interlayer dielectric layer **3** of one or more semiconductor chips may be disposed on (e.g., directly or indirectly beneath) the substrate bottom surface **1a** of the one or more semiconductor chips such that the interlayer dielectric layer **3** of the one or more semiconductor chips is understood to cover the bottom surface **1a** of the substrate **1** of the one or more semiconductor chips. The interlayer dielectric layer **3** may have therein transistors (not shown) and the wiring lines **5** that are multi-layered. The interlayer dielectric layer **3** of one or more semiconductor chips may cover (e.g., partially or completely enclose) wiring lines **5** that are thus understood to be on (e.g., directly or indirectly beneath) the substrate bottom surface **1a** of the one or more semiconductor chips. The interlayer dielectric layer **3** may be covered with the first passivation layer **9** of the one or more semiconductor chips. The first conductive pads **7** of one or more semiconductor chips may be disposed below the interlayer dielectric layer **3** of the one or more semiconductor chips. The conductive bumps **140** of one or more semiconductor chips may be correspondingly bonded to the first conductive pads **7** of the one or more semiconductor chips. A solder layer **150** may be bonded below the

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conductive bumps **140** of the first semiconductor chip **100a**. The substrate top surface **1b** of one or more semiconductor chips may be covered (e.g., directly or indirectly covered) with the second passivation layer **15** of the one or more semiconductor chips. In some example embodiments, the second passivation layer **15** of one or more semiconductor chips may be referred to as a passivation layer that covers a top surface **1b** of the substrate **1** of the one or more semiconductor chips.

Each of the first, second, and third semiconductor chips **100a**, **100b**, and **100c** may further include through vias **11** and through dielectric layers **13**. Each of the first, second, and third semiconductor chips **100a**, **100b**, and **100c** may be configured such that the through via **11** may penetrate the second passivation layer **15**, the substrate **1**, and a portion of the interlayer dielectric layer **3**. The through dielectric layer **13** of one or more semiconductor chips may be interposed between the substrate **1** and the through via **11** of the one or more semiconductor chips. The through via **11** of one or more semiconductor chips may have a top surface in contact with the second conductive pad **21** of the one or more semiconductor chips. For example, a through via **11** of the first semiconductor chip **100a** may penetrate the substrate **1** of the first semiconductor chip **100a** and may be in contact with the second conductive pad **21** of the first semiconductor chip **100a**. The through via **11** of one or more semiconductor chips may have a bottom surface coupled to one of the wiring lines **5** of the one or more semiconductor chips.

The fourth semiconductor chip **100d** may exclude the through via **11** and the through dielectric layer **13**. The substrate **1** included in the fourth semiconductor chip **100d** may have a thickness less (e.g., smaller) than that of the substrate **1** included in either the second semiconductor chip **100b** or the third semiconductor chip **100c**.

The first, second, third, and fourth semiconductor chips **100a**, **100b**, **100c**, and **100d** may be connected to each other through internal connection members **160**. Restated, the semiconductor package **1000** may include internal connection members **160** that lie between adjacent semiconductor chips of the first to fourth semiconductor chips **100a** to **100d** and each connect the adjacent semiconductor chips to each other, such that the internal connection members **160** of the semiconductor package **1000** collectively connect the first to fourth semiconductor chips **100a** to **100d** to each other. Each of the internal connection members **160** may connect the second conductive pad **21** of an underlying one of the first, second, third, and fourth semiconductor chips **100a**, **100b**, **100c**, and **100d** to the conductive bump **140** of a next overlying one of the first, second, third, and fourth semiconductor chips **100a**, **100b**, **100c**, and **100d**. For example, as shown in FIG. 1, an internal connection member **160** may connect the first semiconductor chip **100a** to the second semiconductor chip **100b**, for example based on connecting a second conductive pad **21** of the first semiconductor chip **100a** to a conductive bump **140** of the second semiconductor chip **100b**. As shown, the first semiconductor chip **100a** may include a second conductive pad **21** on the top surface **1b** of the substrate of the first semiconductor chip **100a**, and the second semiconductor chip **100b** may include a conductive bump **140** on the bottom surface **1a** of the substrate **1** of the second semiconductor chip **100b**, and an internal connection member **160** may connect the second conductive pad **21** of the first semiconductor chip **100a** to the conductive bump **140** of the second semiconductor chip **100b**. In some example embodiments, the first conductive pad **7** of the second semiconductor chip **100b** may be between the substrate **1** of the second semiconductor chip **100b** and the

conductive bump **140** of the second semiconductor chip **100b** and further is in contact with the conductive bump **140** of the second semiconductor chip **100b**. The first conductive pad **7** of the second semiconductor chip **100b** may be understood to be adjacent to a bottom surface **1a** of the substrate **1** of the second semiconductor chip **100b**. Under-fill layers **UF** may be interposed between (e.g., directly between) the first, second, third, and fourth semiconductor chips **100a**, **100b**, **100c**, and **100d**.

The substrates **1** may each be a semiconductor substrate, a single-crystalline silicon substrate, or a silicon-on-insulator (SOI) substrate. The interlayer dielectric layers **3** may each include a single or multiple layer including one or more of a silicon oxide layer, a silicon nitride layer, a silicon oxynitride layer, and a porous dielectric layer. The mold layer **MD** may include a dielectric resin, for example, an epoxy molding compound (EMC). The mold layer **MD** may further include filler, which fillers may be dispersed in the dielectric resin. The fillers may include, for example, silicon oxide (SiO_2). The under-fill layer **UF** may include a thermocurable resin or a photo-curable resin. In addition, the under-fill layer **UF** may further include organic fillers or inorganic fillers.

The first conductive pads **7**, the second conductive pads **21**, the conductive bumps **140**, the solder layers **150**, the wiring lines **5**, and through vias **11** may include a conductive material, such as metal. The solder layers **150** may include, for example, SnAg. The wiring lines **5** may independently include, for example, one or more of copper, tungsten, aluminum, ruthenium, titanium, tantalum, titanium nitride, and tantalum nitride. The through vias **11** may include, for example, tungsten.

FIG. 2A illustrates an enlarged view showing section P1 of FIG. 1. FIG. 2B illustrates an enlarged view showing section P2 of FIG. 2A. FIG. 2C illustrates a plan view showing a conductive bump and a second conductive pad according to some example embodiments of the present inventive concepts.

Referring to FIG. 2A, the second passivation layer **15** may include a silicon oxide layer and a silicon nitride layer that are sequentially stacked. The through via **11** and the through dielectric layer **13** may penetrate the second passivation layer **15**. The through dielectric layer **13** may include, for example, a silicon oxide layer. The through via **11** and the through dielectric layer **13** may have their top surfaces coplanar with that of the second passivation layer **15**. The second conductive pad **21** may be on (e.g., indirectly above) the top surface **1b** and may contact the top surface of the through via **11**, the top surface of the through dielectric layer **13**, and a portion of the top surface of the second passivation layer **15**.

The second semiconductor chip **100b** may be configured such that the first conductive pad **7** may contact a bottom surface of the interlayer dielectric layer **3**. The first conductive pad **7** may be connected to one of the wiring lines **5**. The first passivation layer **9** may cover the bottom surface of the interlayer dielectric layer **3**. The first passivation layer **9** may be formed of, for example, a silicon nitride layer, a polyimide layer, or a photo-solder resist (PSR). The first passivation layer **9** may cover a lateral surface of the first conductive pad **7** and a portion of a bottom surface of the first conductive pad **7**. The first passivation layer **9** may partially contact an upper sidewall of the conductive bump **140**. As shown in at least FIG. 2A, the under-fill layer **UF** may contact the top surface of the second passivation layer **15**, a lateral surface of the second conductive pad **21**, a lateral surface of the internal connection member **160**, a

lateral surface of the conductive bump **140**, and a bottom surface of the first passivation layer **9**. Restated, the under-fill layer **UF** may be in simultaneous contact (e.g., direct contact) with a sidewall **21SW** of the second conductive pad **21**, a sidewall **140SW** of the conductive bump **140**, and a sidewall **160SW** of the internal connection member **160**.

The through via **11** may have a first width **W1** in a first direction **X** (which may be referred to herein as a third width in the first direction **X**). The first width **W1** may range, for example, equal to or greater (e.g., larger) than about $4\text{ }\mu\text{m}$. The second conductive pad **21** may have a second width **W2** in the first direction **X** (which may be referred to herein as a first width in the first direction **X**) which may be greater (e.g., larger) than the first width **W1** (e.g., third width) of the through via **11**. For example, the second width **W2** may range from about $4\text{ }\mu\text{m}$ to about $14\text{ }\mu\text{m}$. The conductive bump **140** may have a third width **W3** in the first direction **X** greater (e.g., larger) than the second width **W2**. For example, where the first semiconductor chip **100a** includes a second conductive pad **21** on the top surface **1b** of the substrate **1** of the first semiconductor chip **100a**, and where the second semiconductor chip **100b** includes a conductive bump **140** on the bottom surface **1a** of the substrate **1** of the second semiconductor chip **100b**, the second conductive pad **21** of the first semiconductor chip **100a** may have a second width **W2** in one direction (e.g., the first direction **X** which may be parallel to the top surface **1b** of the first semiconductor chip **100a**) and the conductive bump **140** of the second semiconductor chip **100b** may have a third width **W3** in the one direction, where the second width **W2** is less (e.g., smaller) than the third width **W3**. The first conductive pad **7** of the second semiconductor chip **100b** may have a fourth width **W4** in the first direction **X** that is greater (e.g., larger) than the third width **W3**. The internal connection member may have an inclined sidewall **160SW**. The internal connection member **160** may have a width that increases as approaching the conductive bump **140** from the second conductive pad **21**, or that increases in a second direction **Z**. Restated, the width in the first direction **X** of the internal connection member **160** may narrow with increased proximity to the second conductive pad **21** of the first semiconductor chip **100a** and/or a width of a given cross-sectional area of the internal connection member **160** in the first direction **X** may have a magnitude that is inversely proportional to a distance of said cross-sectional area from the second conductive pad **21** of the first semiconductor chip **100a** in the second direction **Z**. In some example embodiments, the internal connection member **160** may have a width that decreased, or linearly decreases, as approaching the second conductive pad **21** of the first semiconductor chip **100a** from the conductive bump **140** of the second semiconductor chip **100b**. For example, the second width **W2** may be about 0.8 to 0.9 times the third width **W3**.

Referring to FIG. 2C, each of the conductive bump **140** and the second conductive pad **21** of one or more semiconductor chips may have a circular shape when viewed in plan. The second conductive pad **21** may have a top or bottom surface whose area is about 0.64 to 0.81 times that of a top or bottom surface of the conductive bump **140**.

Each of the first, second, and third semiconductor chips **100a**, **100b**, and **100c** may be configured such that the second conductive pad **21** of one or more semiconductor chips may have a width less (e.g., smaller) than that of the conductive bump **140** of one or more semiconductor chips (e.g., the same semiconductor chip, adjacent separate semiconductor chips, all semiconductor chips, etc.). For example, the first semiconductor chip **100a** may be config-

ured such that the second conductive pad **21** of the first semiconductor chip **100a** may have a width less (e.g., smaller) than that of the conductive bump **140** of the first semiconductor chip **100a** and/or of the second semiconductor chip **100b**.

The first conductive pad **7** of one or more semiconductor chips may include, for example, aluminum. In some example embodiments, the second conductive pad **21** of one or more semiconductor chips and the conductive bump **140** of one or more semiconductor chips (e.g., the second conductive pad **21** of the first semiconductor chip **100a** and the conductive bump **140** of the second semiconductor chip **100b**) may include the same first metal. The second conductive pad **21** of one or more semiconductor chips (e.g., the second conductive pad **21** of the first semiconductor chip **100a**) may have a first thickness **T1** in the second direction **Z**. The conductive bump **140** of one or more semiconductor chips (e.g., the conductive bump **140** of the second semiconductor chip **100b**) may have a second thickness **T2** in the second direction **Z** greater (e.g., larger) than the first thickness **T1**.

The internal connection member **160** may include the first metal, and may further include second, third, and fourth metals. In some example embodiments, the internal connection member **160** may include second, third, and fourth metals and may or may not include the first metal. The first to fourth metals may be different from each other. The internal connection member **160** may have first, second, third, fourth, and fifth connection regions **160a**, **160b**, **160c**, **160d**, and **160e** that are sequentially stacked based on distribution of the metals. The first connection region **160a** may contact the second conductive pad **21** (which may be a first conductive pad of the first semiconductor chip **100a**). The fifth connection region **160e** may contact the conductive bump **140** (which may be a second conductive bump of the second semiconductor chip **100b**). The second connection region **160b** and the fourth connection region **160d** may each be spaced apart from (e.g., isolated from direct contact with) the second conductive pad **21** and the conductive bump **140**. In some example embodiments, the internal connection member **160** may have a center that overlaps a center of the through via **11** of the first semiconductor chip **100a**.

The first connection region **160a** and the fifth connection region **160e** may each include the first metal and the second metal, but may each exclude both the third metal and the fourth metal (e.g., may not include either of the third metal or the fourth metal). The third connection region **160c** may include the second metal and the third metal, but may exclude both the first metal and the fourth metal (e.g., may not include either of the first metal or the fourth metal). The second connection region **160b** and the fourth connection region **160d** may each include the first metal, the second metal, and the fourth metal, but may each exclude (e.g., may not include) the third metal. For example, the first metal may be nickel (Ni), the second metal may be tin (Sn), the third metal may be silver (Ag), and the fourth metal may be gold (Au). In this case, the second connection region **160b** and the fourth connection region **160d** may all include gold, and the first, third, and fifth connection regions **160a**, **160c**, and **160e** may all exclude gold. In addition, the second conductive pad **21** and the conductive bump **140** may exclude gold.

For example, the first connection region **160a** and the fifth connection region **160e** may each include Ni_3Sn_4 . The third connection region **160c** may include $\text{Sn}_{1.8}\text{Ag}$. The second connection region **160b** and the fourth connection region **160d** may each include (Au, Ni) Sn_4 .

Referring to FIGS. **2A** and **2B**, the first to fifth connection regions **160a** to **160e** may have therebetween boundaries that are not flat, but are changed depending on their position. For example, the second connection region **160b** may have third thicknesses **T3_i** that are changed depending on a position **X**. The fourth connection region **160d** may have fourth thicknesses **T4_i** that are changed depending on a position **X**. The second connection region **160b** may have a third average thickness **T3** (which may be referred to herein as a first average thickness of the second connection region **160b**), or an average of the third thicknesses **T3_i**. The fourth connection region **160d** may have a fourth average thickness **T4** (which may be referred to herein as a second average thickness of the fourth connection region **160d**), or an average of the fourth thicknesses **T4_i**. For example, the third average thickness **T3** may be greater (e.g., larger) than the fourth average thickness **T4**.

The internal connection members **160** between the first to fourth semiconductor chips **100a** to **100d** shown in FIG. **1** may each have a structure the same as or similar to that discussed with reference to FIGS. **2A** and **2B**. The internal connection member **160** may include therein no void or crack, and thus the semiconductor package **1000** may increase in reliability.

FIGS. **3A** and **3B** illustrate cross-sectional views showing a method of fabricating the semiconductor package of FIG. **1**, according to some example embodiments of the present inventive concepts. FIG. **4** illustrates an enlarged view showing section **P1** of FIG. **3A**, according to some example embodiments of the present inventive concepts.

Referring to FIG. **3A**, a wafer structure **WF1** may be prepared. The wafer structure **WF1** may have a plurality of chip regions **R1** and a separation region **SR** therebetween. The separation region **SR** may be a scribe lane region. The wafer structure **WF1** may include a substrate **1**. The substrate **1** may have a substrate top surface **1b** and a substrate bottom surface **1a** that are opposite to each other. Transistors (not shown) may be formed on the substrate bottom surface **1a**, and a portion of an interlayer dielectric layer **3** may be formed to cover the transistors. The portion of the interlayer dielectric layer **3** and the substrate **1** may be etched to form a through hole, and a through via **11** and a through dielectric layer **13** may be formed in the through hole. Wiring lines **5** may be formed to contact the through via **11**, and a remainder of the interlayer dielectric layer **3** may be formed. First conductive pads **7** and a first passivation layer **9** may be formed on the interlayer dielectric layer **3**. Conductive bumps **140** may be formed on the first conductive pads **7**, and solder layers **150** may be bonded to the conductive bumps **140**. The wafer structure **WF1** may be attached through an adhesive layer **BL** to a carrier substrate **CR**.

Referring to FIGS. **3A** and **4**, one or both of grinding and etch-back processes may be performed on the substrate top surface **1b** of the substrate **1**, such that a portion of the substrate **1** may be removed to expose a top surface of the through via **11**, a top surface of the through dielectric layer **13**, and a portion of an upper sidewall of the through dielectric layer **13**. A second passivation layer **15** may be conformally formed on the substrate top surface **1b**, and then one or both of etch-back and polishing processes may be performed to expose the top surface of the through via **11** and the top surface of the through dielectric layer **13**. A second conductive pad **21** and a wetting layer **155** may be sequentially formed on the through via **11**. The second conductive pad **21** and the wetting layer **155** may have the same second width **W2**, and the second conductive pad **21** may be formed to have a sidewall aligned with that of the

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wetting layer 155. The second conductive pad 21 may include a first metal (e.g., nickel). The wetting layer 155 may include, for example, a fourth metal (e.g., gold).

A second semiconductor chip 100b may be positioned on the chip region R1 of the wafer structure WF1. As discussed in FIG. 1, the second semiconductor chip 100b may include a substrate 1, an interlayer dielectric layer 3, wiring lines 5, first conductive pads 7, a first passivation layer 9, conductive bumps 140, and solder layers 150. The conductive bump 140 may include the first metal (e.g., nickel). The solder layer 150 may include a second metal (e.g., tin) and a third metal (e.g., silver). In addition, the second semiconductor chip 100b may further include through vias 11, through dielectric layers 13, a second passivation layer 15, second conductive pads 21, and wetting layers 155.

Referring to FIG. 4, the second width W2 of each of the wetting layer 155 and the second conductive pad 21 on the wafer structure WF1 may be less (e.g., smaller) than a third width of the conductive bump 140 included in the second semiconductor chip 100b. For example, the second width W2 may be about 0.8 to 0.9 times the third width W3. The second width W2 may range, for example, from about 4 μm to about 14 μm . The wetting layer 155 may have a fifth thickness T5. The fifth thickness T5 may range, for example, from about 0.10 μm to about 0.30 μm . Afterwards, the second semiconductor chip 100b may be positioned on the wafer structure WF1 so as to allow the solder layer 150 and the wetting layer 155 to contact each other, and then a thermal compression process may be performed to bond the solder layer 150 to the wetting layer 155 and simultaneously to form an internal connection member 160 of FIG. 2A. In this case, the second metal (e.g., tin) may diffuse from the solder layer 150 into the second conductive pad 21 and the conductive bump 140. In addition, the fourth metal (e.g., gold) may diffuse from the wetting layer 155 into the conductive bump 140. The diffusion of the second metal (e.g., tin) and the fourth metal (e.g., gold) may form first, second, third, fourth, and fifth connection regions 160a, 160b, 160c, 160d, and 160e of FIG. 2A.

In the present inventive concepts, because the fifth thickness T5 of the wetting layer 155 is greater (e.g., larger) than about 0.10 μm , the solder layer 150 and the wetting layer 155 may be satisfactorily bonded to each other without process fail. When the fifth thickness T5 of the wetting layer 155 is less than about 0.10 μm , the wetting layer 155 may be excessively thin to possibly expose a surface of the second conductive pad 21, and accordingly a problem may occur in which the solder layer 150 is not bonded to the wetting layer 155 or the second conductive pad 21.

Referring to FIG. 3B, identically or similarly to that discussed with reference to FIGS. 3A and 4, a third semiconductor chip 100c and a fourth semiconductor chip 100d may be sequentially stacked on the second semiconductor chip 100b, and then thermal compression processes may be sequentially performed. In some example embodiments, second to fourth semiconductor chips 100b to 100d may be sequentially stacked on each of the chip regions R1 of the wafer structure WF1, and then a thermal compression process may be performed a single time to simultaneously form all of internal connection members 160. Under-fill layers UF may be formed between the wafer structure WF1 and the second semiconductor chips 100b and between the second, third, and fourth semiconductor chips 100b, 100c, and 100d. A mold layer MD may be formed to cover the wafer structure WF1 and the second, third, and fourth semiconductor chips 100b, 100c, and 100d.

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Subsequently, referring to FIG. 1, the carrier substrate CR and the adhesive layer BL may be removed. A singulation process using a laser may be performed such that the wafer structure WF1 and the mold layer MD on the separation region SR may be removed to fabricate a plurality of semiconductor packages 1000. A high-temperature storage (HTS) reliability test may be executed after the semiconductor package 1000 is fabricated as discussed above. For example, the HTS reliability test may be performed for about 1000 hours at about 150° C.

In the present inventive concepts, since the wetting layer 155 has the second width W2 relatively less (e.g., smaller) than the third width W3, an overall amount of the fourth metal (e.g., gold) that constitutes the wetting layer 155 may be relatively less (e.g., smaller) than an amount of the second metal (e.g., tin) included in the solder layer 150. Therefore, the second connection region 160b and the fourth connection region 160d may have their relative small average thicknesses T3 and T4, respectively, and the third connection region 160c may have a relatively large thickness. In this case, the third connection region 160c may still remain even after the HTS reliability test on the semiconductor package 1000. Therefore, the third connection region 160c may have therein no void or crack, and accordingly the semiconductor package 1000 may increase in reliability.

In contrast, when the second width W2 of the wetting layer 155 is the same as or greater (e.g., larger) than the third width W3 of the conductive bump 140, the wetting layer 155 may have a large amount of the fourth metal (e.g., gold) included therein. Therefore, the second connection region 160b and the fourth connection region 160d may have their relative large average thicknesses T3 and T4, respectively, and the third connection region 160c may have a relatively small thickness. In this case, after the HTS reliability test on the semiconductor package 1000, most or all of each of the second metal (e.g., tin) and the third metal (e.g., silver) may migrate from the third connection region 160c into the second connection region 160b and the fourth connection region 160d, which migration may form an intermetallic compound. This metal chemical reaction may induce a reduction in volume, and thus the third connection region 160c may be perished or may have a void or crack therein. An open defect may occur in which the semiconductor chips 100a to 100d are not connected to each other, and accordingly the semiconductor package 1000 may decrease in reliability. A reduction in size of the conductive bump 140 may urge such issues to dominantly affect reliability of the semiconductor package 1000.

In the present inventive concepts, while the thickness T5 of the wetting layer 155 is kept equal to or greater (e.g., larger) than a certain value required to ensure process reliability, the width W2 of the wetting layer 155 may become relatively reduced to comparatively decrease an overall amount of the fourth metal (e.g., gold) included in the wetting layer 155, and thus the formation of intermetallic compound including gold may hardly occur to thereby reduce or prevent the internal connection member 160 from suffering from void or crack.

FIG. 5 illustrates a cross-sectional view showing a semiconductor package according to some example embodiments of the present inventive concepts. FIG. 6 illustrates an enlarged view showing section P1 of FIG. 5.

Referring to FIGS. 5 and 6, a semiconductor package 1001 according to some example embodiments may include a fifth semiconductor chip 100e, a fourth semiconductor chip 100d mounted on the fifth semiconductor chip 100e, and a mold layer MD that covers the fourth and fifth

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semiconductor chips **100d** and **100e**. The fourth semiconductor chip **100d** and the fifth semiconductor chip **100e** may independently be one selected from an image sensor chip such as CMOS image sensor (CIS), a flash memory chip, a dynamic random access memory (DRAM) chip, a static random access memory (SRAM) chip, an electrical erasable programmable read only memory (EEPROM) chip, a phase change random access memory (PRAM) chip, a magnetic random access memory (MRAM) chip, and a resistive random access memory (ReRAM) chip.

The fifth semiconductor chip **100e** may include components of the first semiconductor chip **100a** of FIG. 1, and may further include a first redistribution substrate **RD1** disposed on the second passivation layer **15**. The first redistribution substrate **RD1** may include a first dielectric layer **IL1** and a second dielectric layer **IL2** that are sequentially stacked and each of which may be understood to be between the second conductive pad **21** of the fifth semiconductor chip **100e** and the substrate **1** of the fifth semiconductor chip **100e**, as shown in at least FIG. 6. The first dielectric layer **IL1** and the second dielectric layer **IL2** may each include a photo-imageable dielectric (PID) layer. The first dielectric layer **IL1** may contact the second passivation layer **15**.

Referring to FIG. 6, a first redistribution pattern **RP1** may be interposed between (e.g., directly between) the first dielectric layer **IL1** and the second dielectric layer **IL2**. The first redistribution pattern **RP1** may include a via part **VP** that penetrates the first dielectric layer **IL1** to have a connection with the through via **11**, and may also have a line part **LP** interposed between the first dielectric layer **IL1** and the second dielectric layer **IL2**. A first barrier/seed layer **SL1** may be interposed between the first redistribution pattern **RP1** and the first dielectric layer **IL1** and between the first redistribution pattern **RP1** and the through via **11**. A second redistribution pattern **RP2** may be disposed on the second dielectric layer **IL2**. The second redistribution pattern **RP2** may include a via part **VP** that penetrates the second dielectric layer **IL2** to have a connection with the first redistribution pattern **RP1**, and may also include a pad part **PP** that protrudes onto the second dielectric layer **IL2**. A second barrier/seed layer **SL2** may be interposed between the second redistribution pattern **RP2** and the second dielectric layer **IL2**. The first redistribution pattern **RP1** and the second redistribution pattern **RP2** may include, for example, copper. The first barrier/seed layer **SL1** and the second barrier/seed layer **SL2** may each include a barrier layer and a seed layer that are sequentially stacked. The barrier layer may include, for example, one or more of titanium, tantalum, titanium nitride, and tantalum nitride. The seed layer may include, for example, copper.

In some example embodiments, the second dielectric layer **IL2** may be referred to as a first dielectric layer between the second conductive pad **21** and the substrate **1** of the fifth semiconductor chip **100e**, and the second redistribution pattern **RP2** may be referred to as a first redistribution pattern having a via part **VP** that penetrates the first dielectric layer (**IL2**) and a pad part **PP** that protrudes on to the first dielectric layer (**IL2**).

The second conductive pad **21** may be disposed on the second redistribution pattern **RP2** and may be in contact with the pad part **PP** of the second redistribution pattern **RP2**. The pad part **PP** of the second redistribution pattern **RP2** may have a second width **W2** the same as that of the second conductive pad **21**. The pad part **PP** of the second

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redistribution pattern **RP2** may have a sidewall **PPSW** aligned with (e.g., coplanar with) the sidewall **21SW** of the second conductive pad **21**.

Referring back to FIGS. 5 and 6, the fourth semiconductor chip **100d** may include a substrate **1**, an interlayer dielectric layer **3**, wiring lines **5**, a first passivation layer **9**, first conductive pads **7**, and conductive bumps **140**. An internal connection member **160** may connect the conductive bump **140** of the fourth semiconductor chip **100d** to the second conductive pad **21** of the fifth semiconductor chip **100e**. In some example embodiments, the internal connection member **160** may have a center that does not overlap a center of the through via **11** of the fifth semiconductor chip **100e**. An under-fill layer **UF** may be interposed between the fourth semiconductor chip **100d** and the fifth semiconductor chip **100e**. The under-fill layer **UF** may contact a sidewall of the second redistribution pattern **RP2** and a top surface of the second dielectric layer **IL2**. Other configurations may be identical or similar to those discussed with reference to FIGS. 1 to 2B. The first redistribution substrate **RD1** of FIGS. 5 and 6 may also be disposed on each of the semiconductor chips **100a** to **100d** of FIG. 1.

FIG. 7 illustrates a method of fabricating a semiconductor package having the structure of FIG. 6.

Referring to FIGS. 3A and 7, before a second conductive pad **21** and a wetting layer **155** are formed on the substrate top surface **1b** of the wafer structure **WF1**, a first redistribution substrate **RD1** may be formed on the structure of FIG. 3A. The first redistribution substrate **RD1** may have a structure the same as or similar to that discussed with reference to FIG. 6. A second conductive pad **21** and a wetting layer **155** may be formed on the second redistribution pattern **RP2** of the first redistribution substrate **RD1**. A fourth semiconductor chip **100d** may be positioned on the wafer structure **WF1**. A solder layer **150** of the fourth semiconductor chip **100d** may contact the wetting layer **155** of the wafer structure **WF1**, and then a thermal compression process may be performed to form an internal connection member **160** of FIG. 6. Subsequently, as discussed with reference to FIG. 3B, an under-fill layer **UF** and a mold layer **MD** may be formed, and then a singulation process may be performed to fabricate a semiconductor package **1001** of FIG. 5. Other configurations and fabrication processes may be identical or similar to those discussed with reference to FIGS. 3B and 4.

FIG. 8 illustrates a cross-sectional view showing a semiconductor package according to some example embodiments of the present inventive concepts.

Referring to FIG. 8, a semiconductor package **1002** according to some example embodiments may include a second redistribution substrate **RD2**, a fourth semiconductor chip **100d** mounted on the second redistribution substrate **RD2** (e.g., through an internal connection member **160** on the second redistribution substrate **RD2**), and a mold layer **MD** that covers the fourth semiconductor chip **100d** and the second redistribution substrate **RD2**. The second redistribution substrate **RD2** may include first, second, third, and fourth dielectric layers **IL1**, **IL2**, **IL3**, and **IL4** that are sequentially stacked. Each of the first to fourth dielectric layers **IL1** to **IL4** may include a photo-imageable dielectric (PID) layer. An under-bump **UBM** may be disposed in (e.g., within) the first dielectric layer **IL1**. External connection terminals **300** may be bonded to a bottom surface of the under-bumps **UBM**. The external connection terminals **300** may include, for example, one or more of conductive bumps, conductive pillars, and solder balls. As shown, the second dielectric layer **IL2** may cover the first dielectric layer **IL1**.

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A first redistribution pattern RP1 may be disposed between the third dielectric layer IL3 and the second dielectric layer IL2, such that the third dielectric layer IL3 may cover the first redistribution pattern RP1 and the second dielectric layer IL2. A second redistribution pattern RP2 may be disposed between the fourth dielectric layer IL4 and the third dielectric layer IL3. A third redistribution pattern RP3 may be disposed on the fourth dielectric layer IL4. The internal connection member 160 may be an internal connection member 160 according to any of the example embodiments and may include any of the connection regions according to any of the example embodiments.

A first barrier/seed layer SL1 may be interposed between the first redistribution pattern RP1 and the second dielectric layer IL2. A second barrier/seed layer SL2 may be interposed between the second redistribution pattern RP2 and the third dielectric layer IL3. A third barrier/seed layer SL3 may be interposed between the third redistribution pattern RP3 and the fourth dielectric layer IL4.

The first redistribution pattern RP1 and the second redistribution pattern RP2 may each include a via part VP and a line part LP. For example, the second redistribution pattern RP2 may include a first via part VP and a first line part LP, where the first via part VP penetrates the second dielectric layer IL2 and has a connection with the under-bump UBM (e.g., via the first redistribution pattern RP1), and the first line part LP may protrude onto the second dielectric layer IL2 (e.g., onto a top surface thereof). Likewise the second redistribution pattern RP2 of FIG. 6, the third redistribution pattern RP3 may include a via part VP and a pad part PP, where the via part VP may penetrate the third dielectric layer IL3 and may have a connection with the second redistribution pattern RP2, and the pad part PP may protrude onto the third dielectric layer IL3. A second conductive pad 21 may be disposed on the pad part PP of the third redistribution pattern RP3 and thus may be on (e.g., directly on) the third redistribution pattern RP3. The fourth semiconductor chip 100d may include a conductive bump 140 that protrudes outwardly from a bottom surface 100da of the fourth semiconductor device 100d. An internal connection member 160 may connect the conductive bump 140 of the fourth semiconductor chip 100d to the second conductive pad 21 of the second redistribution substrate RD2. As shown in FIG. 8, a width (e.g., in a first direction X parallel to the top surface 1b and/or bottom surface 1a of the substrate 1 of the fourth semiconductor chip 100d) of the second conductive pad 21 of the second redistribution substrate RD2 may be less (e.g., smaller) than a width of the conductive bump 140 of the fourth semiconductor chip 100d (e.g., the width of the second conductive pad 21 may be about 0.8 to 0.9 times the width of the conductive bump 140). The internal connection member 160 may have a structure the same as or similar to that discussed with reference of FIGS. 2A and 2B. An under-fill layer UF may be interposed between the second redistribution substrate RD2 and the fourth semiconductor chip 100d. The under-fill layer UF may contact a sidewall of the third redistribution pattern RP3 and a top surface of the fourth dielectric layer IL4. Other configurations may be identical or similar to those discussed above with reference to FIG. 5.

FIG. 9 illustrates a cross-sectional view showing a semiconductor package according to some example embodiments of the present inventive concepts.

Referring to FIG. 9, a semiconductor package 1003 according to some example embodiments may include a first sub-semiconductor package PK1 and a second sub-semiconductor package PK2 stacked thereon. The first sub-semicon-

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ductor package PK1 may include components of the semiconductor package 1002 of FIG. 8, and may further include a third redistribution substrate RD3 disposed on the structure of the semiconductor package 1002 of FIG. 8 and mold vias MV that connect a second redistribution substrate RD2 to the third redistribution substrate RD3. It will be understood that the third redistribution substrate RD3 is on (e.g., directly or indirectly above) the first mold layer MD1.

For example, the second redistribution substrate RD2 may have a structure the same as or similar to that discussed with reference to FIG. 8. The second redistribution substrate RD2 may include third redistribution patterns RP3a and RP3b, which third redistribution patterns RP3a and RP3b may include third central redistribution patterns RP3a disposed on a center of the second redistribution substrate RD2 and third edge redistribution patterns RP3b disposed on an edge of the second redistribution substrate RD2. Second central conductive pads 21a may be correspondingly disposed on the third central redistribution patterns RP3a. Second edge conductive pads 21b may be correspondingly disposed on the third edge redistribution patterns RP3b. Wetting layers 155 may be correspondingly disposed on the second edge conductive pads 21b.

A fourth semiconductor chip 100d may be mounted on the second redistribution substrate RD2. First internal connection members 160 may connect the second central conductive pads 21a of the second redistribution substrate RD2 to conductive bumps 140 of the fourth semiconductor chip 100d. One or more of the third redistribution patterns RP3a and RP3b may be disposed on a side of the fourth semiconductor chip 100d. The fourth semiconductor chip 100d and the second redistribution substrate RD2 may be covered with a first mold layer MD1. The mold vias MV may penetrate the first mold layer MD1 and may contact corresponding wetting layers 155 on the second edge conductive pads 21b. The mold vias MV may include, for example, copper.

The third redistribution substrate RD3 may include fifth, sixth, and seventh dielectric layers IL5, IL6, and IL7. The fifth, sixth, and seventh redistribution dielectric layers IL5, IL6, and IL7 may each include a photo-imageable dielectric (PID) layer. A fourth redistribution pattern RP4 may be interposed between the fifth dielectric layer IL5 and the sixth dielectric layer IL6. A fourth barrier/seed layer SL4 may be interposed between the fourth redistribution pattern RP4 and the fifth dielectric layer IL5. The fourth redistribution pattern RP4 may be connected to the mold via MV. A fifth redistribution pattern RP5 may be interposed between the sixth dielectric layer IL6 and the seventh dielectric layer IL7. A fifth barrier/seed layer SL5 may be interposed between the fifth redistribution pattern RP5 and the sixth dielectric layer IL6. The fourth redistribution pattern RP4 and the fifth redistribution pattern RP5 may each include a via part VP and a line part LP. For example, the fifth redistribution pattern RP5 may include a via part VP penetrating the sixth dielectric layer IL6 and having a connection with the mold via MV (e.g., via the fourth redistribution pattern RP4) and a line part LP protruding onto the sixth dielectric layer IL6, and the seventh dielectric layer IL7 may cover the sixth dielectric layer IL6 and the fifth redistribution pattern RP5. A sixth redistribution pattern RP6 may be disposed on the seventh dielectric layer IL7. A sixth barrier/seed layer SL6 may be interposed between the sixth redistribution pattern RP6 and the seventh dielectric layer IL7. The sixth redistribution pattern RP6 may have a via part VP and a pad part PP, where the via part VP may penetrate the seventh dielectric layer IL7 and may have a connection with

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the fifth redistribution pattern RP5 and the pad part PP may protrude onto the seventh dielectric layer IL7. A third conductive pad 31 may be disposed on the pad part PP of the sixth redistribution pattern RP6 and thus may be on (e.g., directly or indirectly above) the sixth redistribution pattern RP6. The fourth to sixth redistribution patterns RP4 to RP6 may each include copper. The third conductive pad 31 may include nickel.

The second sub-semiconductor package PK2 (also referred to herein as an upper semiconductor package) may be on (e.g., may be mounted on) the third redistribution substrate RD3 and may include a package substrate SB, a semiconductor device CH (e.g., second semiconductor device) mounted on (e.g., directly or indirectly above) the package substrate SB (e.g., via an adhesive layer AL), and a second mold layer MD2 that covers the semiconductor device CH and the package substrate SB. The semiconductor device CH may be electrically connected to the package substrate SB through, for example, a wire 360. The semiconductor device CH may be a semiconductor die or chip, or may be a semiconductor package including a plurality of semiconductor dies of the same or different types. The semiconductor device CH may be one selected from an image sensor chip such as CMOS image sensor (CIS), a microelectromechanical system (MEMS) device chip, an application specific integrated circuit (ASIC) chip, and a memory device chip such as a flash memory chip, a dynamic random access memory (DRAM) chip, a static random access memory (SRAM) chip, an electrical erasable programmable read only memory (EEPROM) chip, a phase change random access memory (PRAM) chip, a magnetic random access memory (MRAM) chip, a resistive random access memory (ReRAM) chip, a high bandwidth memory (HBM) chip, and a hybrid memory cubic (HMC) chip.

The second mold layer MD2 may include the same material as that of the first mold layer MD1. The wire 360 may include copper or gold. The package substrate SB may be, for example, a bi-layered or multi-layered printed circuit board. As shown in at least FIG. 9, the package substrate SB may include an upper substrate pad 380 disposed on and/or in a top surface thereof and a lower substrate pad 382 disposed on (e.g., directly or indirectly below) and/or in a bottom surface thereof. As shown in at least FIG. 9, a width of the third conductive pad 31 may be smaller than a width of the lower substrate pad 382. The package substrate SB may have therein an internal line (not shown) that connects the upper substrate pad 380 to the lower substrate pad 382. The upper substrate pad 380 and the lower substrate pad 382 may include one or more of gold, copper, aluminum, and nickel.

The first sub-semiconductor package PK1 may be connected through a second internal connection member 260 to the second sub-semiconductor package PK2. The second internal connection member 260 may connect the lower substrate pad 382 to the third conductive pad 31. The third conductive pad 31 may have a width less (e.g., smaller) than that of the lower substrate pad 382. The pad part PP of the sixth redistribution pattern RP6 may have a width less (e.g., smaller) than that of the lower substrate pad 382. The second internal connection member 260 may have a structure the same as or similar to that of the internal connection member 160 discussed with reference to FIGS. 2A and 2B. Other configurations may be the same as or similar to those discussed with reference to FIG. 8. The mold vias MV may penetrate the first mold layer MD1 and connect the second redistribution substrate RD2 to the third redistribution substrate RD3.

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FIG. 10 illustrates a cross-sectional view showing a semiconductor package according to some example embodiments of the present inventive concepts.

Referring to FIG. 10, a semiconductor package 1004 according to some example embodiments may be configured such that the first sub-semiconductor package PK1 may include a second redistribution substrate RD2, a connection substrate 900 and a fourth semiconductor chip 100d that are mounted on the second redistribution substrate RD2, a first mold layer MD1 that covers the second redistribution substrate RD2, the connection substrate 900, and the fourth semiconductor chip 100d, and a third redistribution substrate RD3 on the first mold layer MD1. As shown, the connection substrate 900 may be between the second and third redistribution substrates RD2 and RD3 and may be covered with the first mold layer MD1.

A first under-fill layer UF1 may be interposed between the fourth semiconductor chip 100d and the second redistribution substrate RD2. The connection substrate 900 may include a cavity CV at a center thereof. Restated, the connection substrate 900 may include one or more inner sidewalls 900SW that at least partially define the cavity CV. The fourth semiconductor chip 100d may be disposed in (e.g., within) the cavity CV at least partially defined by the one or more inner sidewalls 900SW. The connection substrate 900 may include a plurality of base layers 910 and a conductive structure 920. The base layers 910 may include a dielectric material. For example, the base layers 910 may include a carbon-based material, a ceramic, or a polymer. The conductive structure 920 may include a connection pad 921, a first connection via 922, a connection line 923, and a second connection via 924.

The connection substrate 900 may be connected to the second redistribution substrate RD2 through a third internal connection member 316. The connection pad 921 may have a width greater (e.g., larger) than that of the second edge conductive pad 21b. The third internal connection member 316 may have a structure the same as or similar to that of the internal connection member 160 discussed with reference to FIGS. 2A and 2B. A second under-fill layer UF2 may be interposed between the connection substrate 900 and the second redistribution substrate RD2. The first mold layer MD1 may fill a space between the fourth semiconductor chip 100d and an inner sidewall of the cavity CV of the connection substrate 900. The second under-fill layer UF2 may include the same material as that of the first under-fill layer UF1.

A subsidiary via 30 may penetrate the first mold layer MD1 and may connect the second connection via 924 of the connection substrate 900 to the third redistribution pattern RP3 of the second redistribution substrate RD2. A third under-fill layer UF3 may fill a space between the first sub-semiconductor package PK1 and the second sub-semiconductor package PK2. The third under-fill layer UF3 may include the same material as that of the first under-fill layer UF1. Other configurations may be identical or similar to those discussed above with reference to FIG. 9.

FIG. 11 illustrates a cross-sectional view showing a semiconductor package according to some example embodiments of the present inventive concepts.

Referring to FIG. 11, a semiconductor package 2000 according to some example embodiments may be configured such that a second package substrate SB2 may be stacked on a first package substrate SB1. A first sub-semiconductor package PK1 and a second sub-semiconductor package PK2 may be stacked side by side on the second package substrate SB2. The first sub-semiconductor package PK1 and the

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second sub-semiconductor package PK2 may be spaced apart from each other to provide an air gap AG therebetween. A thermal radiation member HS may cover the first sub-semiconductor package PK1, the second sub-semiconductor package PK2, the first package substrate SB1, and the second package substrate SB2. A thermal interface material layers TIM may be interposed between the thermal radiation member HS and the first sub-semiconductor package PK1 and between the thermal radiation member HS and the second sub-semiconductor package PK2. The thermal interface material layer TIM may not fill the air gap AG. In some example embodiments, a layer AB may be between the thermal radiation member HS and the first package substrate SB1. The layer AB may be an adhesive layer, a thermal interface material layer TIM, or the like. The thermal radiation member HS may include a high thermal conductivity material, such as metal. The thermal interface material layer TIM may include grease or a thermo-curable resin layer. The thermal interface material layer TIM may further include filler particles dispersed in the thermosetting resin layer. The filler particles may include one or more of silica, alumina, zinc oxide, and boron nitride.

The first package substrate SB1 may be, for example, a bi-layered or multi-layered printed circuit board. The printed circuit board may include a core and conductive patterns disposed on opposite surfaces of the core. The core may include a thermosetting resin such as an epoxy resin, a thermoplastic resin such as polyimide, a resin (e.g., prepreg) in which a thermosetting or thermoplastic resin is impregnated with a stiffener such as glass fiber and inorganic filler, or a photo-curable resin, but the present inventive concepts are not limited thereto. External connection terminals 300 may be bonded to a bottom surface of the first package substrate SB1. The external connection terminals 300 may include, for example, one or more of conductive bumps, conductive pillars, and solder balls.

The second package substrate SB2 may be, for example, a silicon-based interposer substrate. The second package substrate SB2 may include first substrate conductive pads 210a and second substrate conductive pads 210b disposed on a top surface thereof, and may also include third substrate conductive pads 220 disposed on a bottom surface thereof. The first package substrate SB1 and the second package substrate SB2 may have therebetween first internal connection members 302 that connect the first and second package substrates SB1 and SB2 to each other. The first internal connection members 302 may correspondingly contact the third substrate conductive pads 220. A first under-fill layer UF1 may be interposed between the first package substrate SB1 and the second package substrate SB2. The first internal connection members 302 may include, for example, one or more of conductive bumps, conductive pillars, and solder balls.

The first sub-semiconductor package PK1 may have a structure the same as or similar to that of the second sub-semiconductor package PK2 discussed with reference to FIG. 9. The first sub-semiconductor package PK1 may be electrically connected through third internal connection members 304 to the first substrate conductive pads 210a of the second package substrate SB2.

The second sub-semiconductor package PK2 may have a structure the same as or similar to that of the semiconductor package 1000 of FIG. 1. Conductive bumps 140 may be disposed on a bottom surface of the second sub-semiconductor package PK2. Fourth internal connection members 306 may connect the conductive bumps 140 of the second sub-semiconductor package PK2 to the second substrate

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conductive pads 210b of the second package substrate SB2. The fourth internal connection member 306 may have a structure the same as or similar to that of the internal connection member 160 discussed with reference to FIG. 2A. The second substrate conductive pad 210b may have a width less (e.g., smaller) than that of the conductive bump 140 of the second sub-semiconductor package PK2. A third under-fill layer UF3 may be interposed between the second sub-semiconductor package PK2 and the second package substrate SB2. Conductive pads 210a and 220, conductive pads 210a and 210b, and/or conductive pads 210b and 220 may be connected via one or more conductive patterns and/or wiring 400 extending through the second package substrate SB2.

No void or crack may be present in an internal connection member included in a semiconductor package according to the present inventive concepts, and thus the semiconductor package may increase in reliability.

Although the present inventive concepts have been described in connection with some example embodiments of the present inventive concepts illustrated in the accompanying drawings, it will be understood to those skilled in the art that various changes and modifications may be made without departing from the technical spirit and essential feature of the present inventive concepts. It will be apparent to those skilled in the art that various substitution, modifications, and changes may be thereto without departing from the scope and spirit of the present inventive concepts. The embodiments of FIGS. 1 to 11 may be combined with each other.

What is claimed is:

1. A semiconductor package, comprising:
 - a first redistribution substrate;
 - a first semiconductor device mounted through a first internal connection member on the first redistribution substrate; and
 - a first mold layer that covers the first semiconductor device and the first redistribution substrate,
 wherein the first redistribution substrate includes
 - a plurality of dielectric layers,
 - a plurality of redistribution patterns penetrating at least one of the plurality of dielectric layers, and
 - a first conductive pad above the plurality of redistribution patterns,
 wherein the first semiconductor device includes a conductive bump that protrudes outwardly from a bottom surface of the first semiconductor device, wherein the first internal connection member connects the first conductive pad to the conductive bump, wherein a width of the first conductive pad is smaller than a width of the conductive bump, and wherein the first internal connection member is between the first conductive pad and the conductive bump.
2. The semiconductor package of claim 1, wherein the first redistribution substrate includes:
 - a first dielectric layer,
 - an under-bump in the first dielectric layer,
 - a second dielectric layer that covers the first dielectric layer,
 - a first redistribution pattern that includes a first via part and a first line part, the first via part penetrating the second dielectric layer and having a connection with the under-bump, and the first line part protruding onto the second dielectric layer,
 - a third dielectric layer that covers the first redistribution pattern and the second dielectric layer,

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a second redistribution pattern that includes a second via part and a second line part, the second via part penetrating the third dielectric layer and having a connection with the first redistribution pattern, and the second line part protruding onto the third dielectric layer, 5

a fourth dielectric layer that covers the second redistribution pattern and the third dielectric layer, and

a third redistribution pattern that includes a third via part and a first pad part, the third via part penetrating the fourth dielectric layer. 10

3. The semiconductor package of claim 2, wherein the first conductive pad is disposed on the third redistribution pattern.

4. The semiconductor package of claim 1, further comprising: 15

an under-fill layer between the first redistribution substrate and the first semiconductor device, wherein the under-fill layer is in contact with a sidewall of the first conductive pad.

5. The semiconductor package of claim 1, wherein the 20

width of the first conductive pad is about 0.8 to 0.9 times the width of the conductive bump.

6. The semiconductor package of claim 1, wherein the first internal connection member includes: 25

a first connection region adjacent to the first conductive pad; and

a second connection region adjacent to the conductive bump,

wherein each of the first connection region and the second connection region includes gold (Au), and 30

wherein an average thickness of the first connection region is smaller than an average thickness of the second connection region.

7. The semiconductor package of claim 1, further comprising: 35

a second redistribution substrate on the first mold layer; an upper semiconductor package mounted on the second redistribution substrate;

a second internal connection member that connects the second redistribution substrate to the upper semiconductor package; and 40

a mold via that penetrates the first mold layer and connects the first redistribution substrate to the second redistribution substrate.

8. The semiconductor package of claim 1, further comprising: 45

a second redistribution substrate on the first mold layer; a connection substrate between the first redistribution substrate and the second redistribution substrate and covered with the first mold layer, the connection substrate including a cavity into which the first semiconductor device is inserted; 50

an upper semiconductor package mounted on the second redistribution substrate; and

a second internal connection member that connects the second redistribution substrate to the upper semiconductor package. 55

9. The semiconductor package of claim 7, wherein the second redistribution substrate includes:

a fifth dielectric layer; 60

a fourth redistribution pattern that includes a fourth via part and a third line part, the fourth via part penetrating the fifth dielectric layer and having a connection with the mold via, and the third line part protruding onto the fifth dielectric layer;

a sixth dielectric layer that covers the fifth dielectric layer and the fourth redistribution pattern; 65

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a fifth redistribution pattern that includes a fifth via part and a fourth line part, the fifth via part penetrating the sixth dielectric layer and having a connection with the fourth redistribution pattern, and the fourth line part protruding onto the sixth dielectric layer;

a seventh dielectric layer that covers the sixth dielectric layer and the fifth redistribution pattern;

a sixth redistribution pattern that includes a sixth via part and a second pad part, the sixth via part penetrating the seventh dielectric layer; and

a second conductive pad on the sixth redistribution pattern, 5

wherein the upper semiconductor package includes a package substrate and a second semiconductor device mounted on the package substrate, the package substrate including a substrate pad on a bottom surface of the package substrate,

wherein a width of the second conductive pad is smaller than a width of the substrate pad. 10

10. A semiconductor package, comprising:

a first semiconductor package;

a second semiconductor package on the first semiconductor package; and

a first internal connection member between the first semiconductor package and the second semiconductor package, 15

wherein the first semiconductor package includes

a first redistribution substrate,

a first semiconductor device mounted through a second internal connection member on the first redistribution substrate,

a first mold layer that covers the first semiconductor device and the first redistribution substrate,

a second redistribution substrate on the first mold layer, and 20

a first conductive pad on the second redistribution substrate,

wherein the second semiconductor package includes

a package substrate including an upper substrate pad and a lower substrate pad,

a second semiconductor device connected to the package substrate, and 25

a second molding covering the second semiconductor device,

wherein the first internal connection member is in contact with the first conductive pad of the first semiconductor package and the lower substrate pad of the second semiconductor package,

wherein a width of the first conductive pad is smaller than a width of the lower substrate pad, and 30

wherein the width of the first conductive pad is smaller than or same as a maximum width of the first internal connection member.

11. The semiconductor package of claim 10, further comprising:

a mold via that penetrates the first mold layer and connects the first redistribution substrate to the second redistribution substrate. 35

12. The semiconductor package of claim 10, further comprising:

a wire connecting the package substrate and the second semiconductor device. 40

13. The semiconductor package of claim 10, further comprising:

an under-fill layer between the first semiconductor package and the second semiconductor package, 45

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wherein the under-fill layer is in contact with a sidewall of the first conductive pad.

14. The semiconductor package of claim **10**, further comprising:

a connection substrate between the first and second redistribution substrates and covered with the first mold layer,

wherein the connection substrate includes a plurality of base layers and a conductive structure.

15. A semiconductor package, comprising:

a first package substrate;

a second package substrate on the first package substrate and including a conductive pad;

a first semiconductor package on the second package substrate;

a second semiconductor package on the second package substrate and including a conductive bump disposed on a bottom surface; and

an internal connection member that contacts the conductive bump and the conductive pad,

wherein the second semiconductor package is spaced apart from the first semiconductor package,

wherein a width of the conductive pad is smaller than a width of the conductive bump, and

wherein the width of the conductive pad is smaller than or same as a maximum width of the internal connection member.

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16. The semiconductor package of claim **15**, wherein a width of the internal connection member narrows with increased proximity to the conductive pad.

17. The semiconductor package of claim **15**, further comprising:

an air gap between the first semiconductor package and the second semiconductor package.

18. The semiconductor package of claim **15**, further comprising:

a thermal radiation member covering the first semiconductor package, the second semiconductor package, the first package substrate and the second package substrate.

19. The semiconductor package of claim **18**, further comprising:

a thermal interface material layer interposed between the thermal radiation member and the first semiconductor package, and the thermal radiation member and the second semiconductor package.

20. The semiconductor package of claim **15**, further comprising:

an under-fill layer between the first semiconductor package and the second package substrate, the second semiconductor package and the second package substrate, and the first package substrate and the second package substrate.

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